Human review packet: EOS07GSP

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: EOS07GSP
Paper id: `EOS07GSP`
Source version: NBER w11765; byte-pinned EOS07GSP.txt source extraction
Source record: `audit/paper_statement_map.json`
Rows: 24
Generated: 2026-08-18
Source URL: official source

Review status: 0/24 source-claim screens; 0/140 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper EOS07GSP --serve
```

Open `http://127.0.0.1:8765/` in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate.clickThroughRate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) → (a : ℕ) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate.click_pos`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) (rank : ℕ),
  0 < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem7_bstar_bid`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(value vcgTotalPayment clickThroughRate : ℕ → ℝ) →
  (rank : ℕ) →
    match rank with
    | 0 => value 0
    | i.succ => vcgTotalPayment i / clickThroughRate i
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(value clickThroughRate : ℕ → ℝ) →
  (a a_1 : ℕ) →
    Nat.brecOn (motive := fun x => ℕ → ℝ) a_1
      (fun x f x_1 =>
        (match (motive := ℕ → (x : ℕ) → Nat.below (motive := fun x => ℕ → ℝ) x → ℝ) x_1, x with
          | _i, 0 => fun x => 0
          | i, remaining.succ => fun x => (clickThroughRate i - clickThroughRate (i + 1)) * value (i + 1) + x.1 (i + 1))
          f)
      a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
(clickThroughRate value : ℕ → ℝ) →
  (remaining : ℕ) →
    (∀ (rank : ℕ), 0 < clickThroughRate rank) →
      (∀ (rank : ℕ), 0 ≤ clickThroughRate (rank + 1)) →
        (∀ (rank : ℕ), clickThroughRate (rank + 1) ≤ clickThroughRate rank) →
          (∀ (rank : ℕ),
              EconCSLib.Auction.paper_theorem7_bstar_bid value
                  (fun j => EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate j remaining)
                  clickThroughRate (rank + 2) ≤
                value (rank + 1)) →
            EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate.remaining`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) → self.3
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate.value`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) → (a : ℕ) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate.clickThroughRate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) → (a : ℕ) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate.click_pos`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) (rank : ℕ),
  0 < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
(clickThroughRate value : ℕ → ℝ) →
  (remaining : ℕ) →
    (∀ (rank : ℕ), 0 < clickThroughRate rank) →
      (∀ (rank : ℕ), clickThroughRate (rank + 1) < clickThroughRate rank) →
        (∀ (rank : ℕ),
            EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate (rank + 1) remaining <
              clickThroughRate (rank + 1) * value (rank + 1)) →
          EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate.remaining`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) → self.3
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate.value`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) → (a : ℕ) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertificate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.clockPrice

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} → (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) → self.1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_2 → Type u_3 → Type u_4 → Type (max (max (max u_1 u_2) u_3) u_4)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.isConsistentBelief`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} {StrategyProfile : Type u_3} {Belief : Type u_4}
  (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief)
  (a : StrategyProfile) (a_1 : Belief), self.3 a a_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.isSequentiallyRational

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} {StrategyProfile : Type u_3} {Belief : Type u_4}
  (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief)
  (a : StrategyProfile) (a_1 : Belief), self.4 a a_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.PerfectBayesianEquilibrium`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} {StrategyProfile : Type u_3} {Belief : Type u_4}
  (G : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief)
  (strategy : StrategyProfile),
  ∃ belief, G.isConsistentBelief strategy belief ∧ G.isSequentiallyRational strategy belief
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.environment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    {StrategyProfile : Type u_3} →
      {Belief : Type u_4} →
        (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief) →
          self.1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.outcomeOf

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    {StrategyProfile : Type u_3} →
      {Belief : Type u_4} →
        (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief) →
          (a : StrategyProfile) → self.5 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.values

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    {StrategyProfile : Type u_3} →
      {Belief : Type u_4} →
        (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief) →
          (a : Bidder) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.vcgOutcome`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    {StrategyProfile : Type u_3} →
      {Belief : Type u_4} →
        (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief) →
          self.6
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(Bidder : Type u_1) → EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder → Bidder → Prop
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionEnvironment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_1
```

Exact Lean library declaration

```
structure PositionEnvironment (Slot : Type*) where
  clickThroughRate : Slot → ℝ
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionEnvironment.clickThroughRate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Slot : Type u_1} → (self : EconCSLib.Auction.PositionEnvironment Slot) → (a : Slot) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_2 → Type (max u_1 u_2)
```

Exact Lean library declaration

```
structure PositionOutcome (Bidder Slot : Type*) where
  slotOf : Bidder → Option Slot
  paymentPerClick : Bidder → ℝ
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_threshold_bid

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(value clickThroughRate : ℕ → ℝ) →
  (remaining rank : ℕ) →
    EconCSLib.Auction.paper_theorem7_bstar_bid value
      (fun j => EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate j remaining)
      clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_generalized_english_indifference_price

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(alphaAbove alphaCurrent lastDropout value : ℝ) → value - alphaCurrent / alphaAbove * (value - lastDropout)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(clickThroughRate lastDropout value : ℕ → ℝ) →
  (rank : ℕ) →
    EconCSLib.Auction.paper_theorem8_generalized_english_indifference_price (clickThroughRate rank)
      (clickThroughRate (rank + 1)) (lastDropout rank) (value (rank + 1))
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_threshold_dropout_strategy`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} (dropoutPrice : Bidder → ℝ)
  (a : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) (a_1 : Bidder),
  dropoutPrice a_1 ≤ a.clockPrice
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_ranked_threshold_dropout_strategy`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate lastDropout value : ℕ → ℝ) (a : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
  (a_1 : ℕ),
  EconCSLib.Auction.paper_theorem8_threshold_dropout_strategy
    (fun rank =>
      EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout value rank)
    a a_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening   Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_optimality_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining rank : ℕ),
  EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (rank + 2) ≤
      EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1) (rank + 1) ∧
    EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1) (rank + 1) ≤
        value (rank + 1) ∧
      (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
          state.clockPrice <
              EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (rank + 2) →
            ¬EconCSLib.Auction.paper_theorem8_ranked_threshold_dropout_strategy clickThroughRate
                (fun k => EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (k + 2))
                value state rank) ∧
        (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
            value (rank + 1) ≤ state.clockPrice →
              EconCSLib.Auction.paper_theorem8_ranked_threshold_dropout_strategy clickThroughRate
                (fun k => EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (k + 2))
                value state rank) ∧
          (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
              state.clockPrice =
                  EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1)
                    (rank + 1) →
                clickThroughRate rank * (value (rank + 1) - state.clockPrice) =
                  clickThroughRate (rank + 1) *
                    (value (rank + 1) -
                      EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining
                        (rank + 2))) ∧
            (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
                state.clockPrice <
                    EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1)
                      (rank + 1) →
                  clickThroughRate (rank + 1) *
                      (value (rank + 1) -
                        EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining
                          (rank + 2)) <
                    clickThroughRate rank * (value (rank + 1) - state.clockPrice)) ∧
              ∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
                EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1) (rank + 1) <
                    state.clockPrice →
                  clickThroughRate rank * (value (rank + 1) - state.clockPrice) <
                    clickThroughRate (rank + 1) *
                      (value (rank + 1) -
                        EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining
                          (rank + 2))
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.mk`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Bidder : Type u_1} →
  {Slot : Type u_2} → (Bidder → Option Slot) → (Bidder → ℝ) → EconCSLib.Auction.PositionOutcome Bidder Slot
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(value clickThroughRate : ℕ → ℝ) →
  (remaining : ℕ) →
    { slotOf := fun i => some i,
      paymentPerClick := fun i =>
        EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (i + 1) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (value clickThroughRate : ℕ → ℝ) (remaining : ℕ) (a : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
  (a_1 : ℕ),
  EconCSLib.Auction.paper_theorem8_ranked_threshold_dropout_strategy clickThroughRate
    (fun k => EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (k + 2)) value a a_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.
PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertificate.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Belief : Type u_1} →
  (game :
      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame ℕ ℕ
        (EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ) Belief) →
    (strictModel : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) →
      (∀ (rank : ℕ), game.environment.clickThroughRate rank = strictModel.clickThroughRate rank) →
        (∀ (rank : ℕ), game.values rank = strictModel.value rank) →
          (belief : Belief) →
            game.isConsistentBelief
                (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy strictModel.value
                  strictModel.clickThroughRate strictModel.remaining)
                belief →
              (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_optimality_statement
                    strictModel.clickThroughRate strictModel.value strictModel.remaining →
                  game.isSequentiallyRational
                    (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy strictModel.value
                      strictModel.clickThroughRate strictModel.remaining)
                    belief) →
                (∀ (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ),
                    game.PerfectBayesianEquilibrium strategy →
                      ∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) (rank : ℕ),
                        strategy state rank ↔
                          EconCSLib.Auction.paper_theorem8_bstar_threshold_bid strictModel.value
                              strictModel.clickThroughRate (strictModel.remaining + 1) (rank + 1) ≤
                            state.clockPrice) →
                  game.outcomeOf
                        (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy strictModel.value
                          strictModel.clickThroughRate strictModel.remaining) =
                      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome strictModel.value
                        strictModel.clickThroughRate strictModel.remaining →
                    game.vcgOutcome =
                        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome strictModel.value
                          strictModel.clickThroughRate strictModel.remaining →
                      EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertif ⌟
                      ↪ icate
                        Belief
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertificate.strictModel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Belief : Type u_1} →
  (self :
      EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertificate
        Belief) →
    self.2
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.
PaperTheorem8BStarRankedThresholdStrictOrderedDynamicGameConstructedOutcomeCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.
PaperTheorem8BStarRankedThresholdStrictOrderedDynamicGameConstructedOutcomeCertificate.base

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Belief : Type u_1} →
  (self :
      EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedDynamicGameConstructedOutcomeCertificate Belief) →
    self.1
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.`
`PaperTheorem8BStarRankedThresholdStrictOrderedDynamicGameConstructedOutcomeCertificate.mk`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Belief : Type u_1} →
  (base :
      EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityDynamicGameConstructedOutcomeCertificate
        Belief) →
    (∀ (rank : ℕ), 0 ≤ base.strictModel.value rank) →
      (∀ (rank : ℕ), base.strictModel.value (rank + 1) ≤ base.strictModel.value rank) →
        EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedDynamicGameConstructedOutcomeCertificate Belief
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.
clickThroughRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) → (a : ℕ) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.click_pos

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) (rank : ℕ),
  0 < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.remaining`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) → self.3
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.value`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) → (a : ℕ) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.continuation_tail_payment_lt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) (rank : ℕ),
  EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment self.value self.clickThroughRate (rank + 1) self.remaining <
    self.clickThroughRate (rank + 1) * self.value (rank + 1)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.current_lt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) (rank : ℕ),
  self.clickThroughRate (rank + 1) < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.mk`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
(clickThroughRate value : ℕ → ℝ) →
  (remaining : ℕ) →
    (∀ (rank : ℕ), 0 < clickThroughRate rank) →
      (∀ (rank : ℕ), clickThroughRate (rank + 1) < clickThroughRate rank) →
        (∀ (rank : ℕ), 0 ≤ value rank) →
          (∀ (rank : ℕ), value (rank + 1) ≤ value rank) →
            (∀ (rank : ℕ),
                EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate (rank + 1) remaining <
                  clickThroughRate (rank + 1) * value (rank + 1)) →
              EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.value_mono

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) (rank : ℕ),
  self.value (rank + 1) ≤ self.value rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate.value_nonneg

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) (rank : ℕ),
  0 ≤ self.value rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.clickThroughRate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) → (a : ℕ) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.click_pos`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) (rank : ℕ),
  0 < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.current_lt

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) (rank : ℕ),
  self.clickThroughRate (rank + 1) < self.clickThroughRate rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.remaining`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) → self.3
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.value`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) → (a : ℕ) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate.value_nonneg

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (self : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) (rank : ℕ),
  0 ≤ self.value rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.lastDropout`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  (self : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) → (a : Bidder) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.IsActive

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) (bidder : Bidder),
  state.lastDropout bidder = none
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyHistory

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Bidder : Type u_1} →
  [DecidableEq Bidder] →
    (EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder → Bidder → Prop) →
      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder →
        EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder → Prop
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyTerminal`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder → Bidder → Prop)
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) (bidder : Bidder),
  state.IsActive bidder → ¬strategy state bidder
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Bidder : Type u_1} → ℝ → (Bidder → Option ℝ) → EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.advanceClock`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) →
    (newPrice : ℝ) → { clockPrice := newPrice, lastDropout := state.lastDropout }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.recordDropout

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  [inst : DecidableEq Bidder] →
    (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState Bidder) →
      (bidder : Bidder) →
        { clockPrice := state.clockPrice,
          lastDropout := fun j => if j = bidder then some state.clockPrice else state.lastDropout j }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    {StrategyProfile : Type u_3} →
      {Belief : Type u_4} →
        EconCSLib.Auction.PositionEnvironment Slot →
          (Bidder → ℝ) →
            (StrategyProfile → Belief → Prop) →
              (StrategyProfile → Belief → Prop) →
                (StrategyProfile → EconCSLib.Auction.PositionOutcome Bidder Slot) →
                  EconCSLib.Auction.PositionOutcome Bidder Slot →
                    EconCSLib.Auction.PaperTheorem8GeneralizedEnglishDynamicGame Bidder Slot StrategyProfile Belief
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionEnvironment.mk`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Slot : Type u_1} → (Slot → ℝ) → EconCSLib.Auction.PositionEnvironment Slot
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(Bidder : Type u_1) → (Slot : Type u_2) → (Bidder → ℝ) → EconCSLib.Auction.PositionOutcome Bidder Slot
```

Exact Lean library declaration

```
abbrev PositionMechanism (Bidder Slot : Type*) :=
  (Bidder → ℝ) → PositionOutcome Bidder Slot
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.paymentPerClick`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} → (self : EconCSLib.Auction.PositionOutcome Bidder Slot) → (a : Bidder) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.slotOf`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} → (self : EconCSLib.Auction.PositionOutcome Bidder Slot) → (a : Bidder) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.utility

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    (E : EconCSLib.Auction.PositionEnvironment Slot) →
      (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
        (values : Bidder → ℝ) →
          (i : Bidder) →
            match O.slotOf i with
            | none => 0
            | some s => E.clickThroughRate s * (values i - O.paymentPerClick i)
```

Exact Lean library declaration

```
def utility (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot)
    (values : Bidder → ℝ) (i : Bidder) : ℝ :=
  match O.slotOf i with
  | none => 0
  | some s => E.clickThroughRate s * (values i - O.paymentPerClick i)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.utility`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    (E : EconCSLib.Auction.PositionEnvironment Slot) →
      (M : EconCSLib.Auction.PositionMechanism Bidder Slot) →
        (values bids : Bidder → ℝ) → (i : Bidder) → EconCSLib.Auction.PositionOutcome.utility E (M bids) values i
```

Exact Lean library declaration

```
def utility (E : PositionEnvironment Slot)
    (M : PositionMechanism Bidder Slot)
    (values bids : Bidder → ℝ) (i : Bidder) : ℝ :=
  (M bids).utility E values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.IsNashEquilibrium`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : DecidableEq Bidder] (E : EconCSLib.Auction.PositionEnvironment Slot)
  (M : EconCSLib.Auction.PositionMechanism Bidder Slot) (values bids : Bidder → ℝ) (i : Bidder) (report : ℝ),
  EconCSLib.Auction.PositionMechanism.utility E M values (Function.update bids i report) i ≤
    EconCSLib.Auction.PositionMechanism.utility E M values bids i
```

Exact Lean library declaration

```
def IsNashEquilibrium [DecidableEq Bidder]
    (E : PositionEnvironment Slot) (M : PositionMechanism Bidder Slot)
    (values bids : Bidder → ℝ) : Prop :=
  ∀ (i : Bidder) (report : ℝ),
    utility E M values (Function.update bids i report) i ≤
      utility E M values bids i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.SlotEnvyFree

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (E : EconCSLib.Auction.PositionEnvironment Slot)
  (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (values : Bidder → ℝ) (i j : Bidder) (s : Slot),
  O.slotOf j = some s →
    E.clickThroughRate s * (values i - O.paymentPerClick j) ≤ EconCSLib.Auction.PositionOutcome.utility E O values i
```

Exact Lean library declaration

```
def SlotEnvyFree (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ) : Prop :=
  ∀ (i j : Bidder) (s : Slot),
    O.slotOf j = some s →
      E.clickThroughRate s * (values i - O.paymentPerClick j) ≤
        O.utility E values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.LocallyEnvyFreeEquilibrium`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : DecidableEq Bidder] (E : EconCSLib.Auction.PositionEnvironment Slot)
  (M : EconCSLib.Auction.PositionMechanism Bidder Slot) (values bids : Bidder → ℝ),
  EconCSLib.Auction.PositionMechanism.IsNashEquilibrium E M values bids ∧
    EconCSLib.Auction.PositionOutcome.SlotEnvyFree E (M bids) values
```

Exact Lean library declaration

```
def LocallyEnvyFreeEquilibrium [DecidableEq Bidder]
    (E : PositionEnvironment Slot) (M : PositionMechanism Bidder Slot)
    (values bids : Bidder → ℝ) : Prop :=
  M.IsNashEquilibrium E values bids ∧
    (M bids).SlotEnvyFree E values
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionMechanism.TruthfulDominantStrategy

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : DecidableEq Bidder] (E : EconCSLib.Auction.PositionEnvironment Slot)
  (M : EconCSLib.Auction.PositionMechanism Bidder Slot) (values : Bidder → ℝ) (i : Bidder) (report : ℝ),
  EconCSLib.Auction.PositionMechanism.utility E M values (Function.update values i report) i ≤
    EconCSLib.Auction.PositionMechanism.utility E M values values i
```

Exact Lean library declaration

```
def TruthfulDominantStrategy [DecidableEq Bidder]
    (E : PositionEnvironment Slot) (M : PositionMechanism Bidder Slot) : Prop :=
  ∀ (values : Bidder → ℝ) (i : Bidder) (report : ℝ),
    utility E M values (Function.update values i report) i ≤
      utility E M values values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.FeasibleAssignment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (O : EconCSLib.Auction.PositionOutcome Bidder Slot) {{i j : Bidder}} {{s : Slot}},
  O.slotOf i = some s → O.slotOf j = some s → i = j
```

Exact Lean library declaration

```
def FeasibleAssignment (O : PositionOutcome Bidder Slot) : Prop :=
  ∀ {{i j : Bidder}} {{s : Slot}},
    O.slotOf i = some s → O.slotOf j = some s → i = j
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.welfare

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      (E : EconCSLib.Auction.PositionEnvironment Slot) →
        (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
          (values : Bidder → ℝ) →
            ∑ i,
              match O.slotOf i with
              | none => 0
              | some s => E.clickThroughRate s * values i
```

Exact Lean library declaration

```
noncomputable def welfare [Fintype Bidder]
    (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ) : ℝ :=
  ∑ i : Bidder,
    match O.slotOf i with
    | none => 0
    | some s => E.clickThroughRate s * values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} → (slotOf : Bidder → Option Slot) → { slotOf := slotOf, paymentPerClick := fun x => 0 }
```

Exact Lean library declaration

```
def zeroPaymentOutcome (slotOf : Bidder → Option Slot) :
    PositionOutcome Bidder Slot where
  slotOf := slotOf
  paymentPerClick := fun _ => 0
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionMechanism.exists_welfareMaximizingSlotAssignment

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : Fintype Bidder] [DecidableEq Bidder] [Fintype Slot] [DecidableEq Slot]
  (E : EconCSLib.Auction.PositionEnvironment Slot) (reports : Bidder → ℝ),
  ∃ slotOf,
    (EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome slotOf).FeasibleAssignment ∧
      ∀ (otherSlotOf : Bidder → Option Slot),
        (EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome otherSlotOf).FeasibleAssignment →
          EconCSLib.Auction.PositionOutcome.welfare E (EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome otherSlotOf)
              reports ≤
            EconCSLib.Auction.PositionOutcome.welfare E (EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome slotOf)
              reports
```

Exact Lean library declaration

```
theorem exists_welfareMaximizingSlotAssignment
    [Fintype Bidder] [DecidableEq Bidder]
    [Fintype Slot] [DecidableEq Slot]
    (E : PositionEnvironment Slot) (reports : Bidder → ℝ) :
    ∃ slotOf : Bidder → Option Slot,
      (PositionOutcome.zeroPaymentOutcome slotOf).FeasibleAssignment ∧
        ∀ otherSlotOf : Bidder → Option Slot,
          (PositionOutcome.zeroPaymentOutcome otherSlotOf).FeasibleAssignment →
            (PositionOutcome.zeroPaymentOutcome otherSlotOf).welfare E reports ≤
              (PositionOutcome.zeroPaymentOutcome slotOf).welfare E reports
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.reportsWithoutBidder`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  [inst : DecidableEq Bidder] → (reports : Bidder → ℝ) → (i a : Bidder) → Function.update reports i 0 a
```

Exact Lean library declaration

```
def reportsWithoutBidder [DecidableEq Bidder]
    (reports : Bidder → ℝ) (i : Bidder) : Bidder → ℝ :=
  Function.update reports i 0
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.welfareContribution`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    (E : EconCSLib.Auction.PositionEnvironment Slot) →
      (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
        (values : Bidder → ℝ) →
          (i : Bidder) →
            match O.slotOf i with
            | none => 0
            | some s => E.clickThroughRate s * values i
```

Exact Lean library declaration

```
def welfareContribution
    (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ)
    (i : Bidder) : ℝ :=
  match O.slotOf i with
  | none => 0
  | some s => E.clickThroughRate s * values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.welfareExcept

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      [inst_1 : DecidableEq Bidder] →
        (E : EconCSLib.Auction.PositionEnvironment Slot) →
          (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
            (values : Bidder → ℝ) →
              (i : Bidder) →
                ∑ j ∈ Finset.univ.erase i, EconCSLib.Auction.PositionOutcome.welfareContribution E O values j
```

Exact Lean library declaration

```
noncomputable def welfareExcept [Fintype Bidder] [DecidableEq Bidder]
    (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ)
    (i : Bidder) : ℝ :=
  (Finset.univ.erase i).sum fun j => O.welfareContribution E values j
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionMechanism.positionVCGBenchmark

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      [inst_1 : DecidableEq Bidder] →
        (E : EconCSLib.Auction.PositionEnvironment Slot) →
          (slotRule : (Bidder → ℝ) → Bidder → Option Slot) →
            (reports : Bidder → ℝ) →
              (i : Bidder) →
                EconCSLib.Auction.PositionOutcome.welfareExcept E
                  (EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome
                    (slotRule (EconCSLib.Auction.PositionMechanism.reportsWithoutBidder reports i)))
                  reports i
```

Exact Lean library declaration

```
noncomputable def positionVCGBenchmark
    [Fintype Bidder] [DecidableEq Bidder]
    (E : PositionEnvironment Slot)
    (slotRule : (Bidder → ℝ) → Bidder → Option Slot)
    (reports : Bidder → ℝ) (i : Bidder) : ℝ :=
  (PositionOutcome.zeroPaymentOutcome
      (slotRule (reportsWithoutBidder reports i))).welfareExcept E reports i
```

Recorded source-to-library screening   **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionMechanism.positionVCGMechanismOfSlotRule

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      [inst_1 : DecidableEq Bidder] →
        (E : EconCSLib.Auction.PositionEnvironment Slot) →
          (slotRule : (Bidder → ℝ) → Bidder → Option Slot) →
            (a : Bidder → ℝ) →
              have chosen := EconCSLib.Auction.PositionOutcome.zeroPaymentOutcome (slotRule a);
              { slotOf := slotRule a,
                paymentPerClick := fun i =>
                  match slotRule a i with
                  | none => 0
                  | some s =>
                    (EconCSLib.Auction.PositionMechanism.positionVCGBenchmark E slotRule a i -
                        EconCSLib.Auction.PositionOutcome.welfareExcept E chosen a i) /
                      E.clickThroughRate s }
```

Exact Lean library declaration

```
noncomputable def positionVCGMechanismOfSlotRule
    [Fintype Bidder] [DecidableEq Bidder]
    (E : PositionEnvironment Slot)
    (slotRule : (Bidder → ℝ) → Bidder → Option Slot) :
    PositionMechanism Bidder Slot :=
  fun reports =>
    let chosen :=
      PositionOutcome.zeroPaymentOutcome (slotRule reports)
    { slotOf := slotRule reports
      paymentPerClick := fun i =>
        match slotRule reports i with
        | none => 0
        | some s =>
            (positionVCGBenchmark E slotRule reports i -
              chosen.welfareExcept E reports i) / E.clickThroughRate s }
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.welfareMaximizingSlotAssignment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      [inst_1 : DecidableEq Bidder] →
        [inst_2 : Fintype Slot] →
          [inst_3 : DecidableEq Slot] →
            (E : EconCSLib.Auction.PositionEnvironment Slot) →
              (reports : Bidder → ℝ) → (a : Bidder) → Classical.choose ⋯ a
```

Exact Lean library declaration

```
noncomputable def welfareMaximizingSlotAssignment
    [Fintype Bidder] [DecidableEq Bidder]
    [Fintype Slot] [DecidableEq Slot]
    (E : PositionEnvironment Slot) (reports : Bidder → ℝ) :
    Bidder → Option Slot :=
  Classical.choose (exists_welfareMaximizingSlotAssignment E reports)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionMechanism.positionVCGMechanism`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      [inst_1 : DecidableEq Bidder] →
        [inst_2 : Fintype Slot] →
          [inst_3 : DecidableEq Slot] →
            (E : EconCSLib.Auction.PositionEnvironment Slot) →
              (a : Bidder → ℝ) →
                EconCSLib.Auction.PositionMechanism.positionVCGMechanismOfSlotRule E
                  (EconCSLib.Auction.PositionMechanism.welfareMaximizingSlotAssignment E) a
```

Exact Lean library declaration

```
noncomputable def positionVCGMechanism
    [Fintype Bidder] [DecidableEq Bidder]
    [Fintype Slot] [DecidableEq Slot]
    (E : PositionEnvironment Slot) :
    PositionMechanism Bidder Slot :=
  positionVCGMechanismOfSlotRule E (welfareMaximizingSlotAssignment E)
```

Recorded source-to-library screening   Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.IndividuallyRational

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (E : EconCSLib.Auction.PositionEnvironment Slot)
  (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (values : Bidder → ℝ) (i : Bidder),
  0 ≤ EconCSLib.Auction.PositionOutcome.utility E O values i
```

Exact Lean library declaration

```
def IndividuallyRational (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ) : Prop :=
  ∀ i, 0 ≤ O.utility E values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.PositionOutcome.NoProfitableAssignedSlotDeviation

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (E : EconCSLib.Auction.PositionEnvironment Slot)
  (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (values : Bidder → ℝ) (i j : Bidder) (s : Slot),
  O.slotOf j = some s →
    E.clickThroughRate s * (values i - O.paymentPerClick j) ≤ EconCSLib.Auction.PositionOutcome.utility E O values i
```

Exact Lean library declaration

```
def NoProfitableAssignedSlotDeviation (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ) : Prop :=
  ∀ (i j : Bidder) (s : Slot),
    O.slotOf j = some s →
      E.clickThroughRate s * (values i - O.paymentPerClick j) ≤
        O.utility E values i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.StableAssignment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (E : EconCSLib.Auction.PositionEnvironment Slot)
  (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (values : Bidder → ℝ),
  O.FeasibleAssignment ∧
    EconCSLib.Auction.PositionOutcome.IndividuallyRational E O values ∧
      EconCSLib.Auction.PositionOutcome.NoProfitableAssignedSlotDeviation E O values
```

Exact Lean library declaration

```
def StableAssignment (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (values : Bidder → ℝ) : Prop :=
  O.FeasibleAssignment ∧
    O.IndividuallyRational E values ∧
      O.NoProfitableAssignedSlotDeviation E values
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.revenueContribution`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    (E : EconCSLib.Auction.PositionEnvironment Slot) →
      (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
        (i : Bidder) →
          match O.slotOf i with
          | none => 0
          | some s => E.clickThroughRate s * O.paymentPerClick i
```

Exact Lean library declaration

```
def revenueContribution (E : PositionEnvironment Slot)
    (O : PositionOutcome Bidder Slot) (i : Bidder) : ℝ :=
  match O.slotOf i with
  | none => 0
  | some s => E.clickThroughRate s * O.paymentPerClick i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.PositionOutcome.revenue`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Bidder : Type u_1} →
  {Slot : Type u_2} →
    [inst : Fintype Bidder] →
      (E : EconCSLib.Auction.PositionEnvironment Slot) →
        (O : EconCSLib.Auction.PositionOutcome Bidder Slot) →
          ∑ i, EconCSLib.Auction.PositionOutcome.revenueContribution E O i
```

Exact Lean library declaration

```
noncomputable def revenue [Fintype Bidder]
    (E : PositionEnvironment Slot) (O : PositionOutcome Bidder Slot) : ℝ :=
  ∑ i : Bidder, revenueContribution E O i
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.secondBidder3

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(bids : Fin 3 → ℝ) →
  if bids 0 < bids 1 then if bids 1 < bids 2 then 1 else if bids 0 < bids 2 then 2 else 0
  else if bids 0 < bids 2 then 0 else if bids 1 < bids 2 then 2 else 1
```

Exact Lean library declaration

```
noncomputable def secondBidder3 (bids : Fin 3 → ℝ) : Fin 3 :=
  if bids (0 : Fin 3) < bids (1 : Fin 3) then
    if bids (1 : Fin 3) < bids (2 : Fin 3) then
      (1 : Fin 3)
    else if bids (0 : Fin 3) < bids (2 : Fin 3) then
      (2 : Fin 3)
    else
      (0 : Fin 3)
  else
    if bids (0 : Fin 3) < bids (2 : Fin 3) then
      (0 : Fin 3)
    else if bids (1 : Fin 3) < bids (2 : Fin 3) then
      (2 : Fin 3)
    else
      (1 : Fin 3)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.thirdBidder3`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(bids : Fin 3 → ℝ) →
  if bids 0 < bids 1 then if bids 1 < bids 2 then 0 else if bids 0 < bids 2 then 0 else 2
  else if bids 0 < bids 2 then 1 else if bids 1 < bids 2 then 1 else 2
```

Exact Lean library declaration

```
noncomputable def thirdBidder3 (bids : Fin 3 → ℝ) : Fin 3 :=
  if bids (0 : Fin 3) < bids (1 : Fin 3) then
    if bids (1 : Fin 3) < bids (2 : Fin 3) then
      (0 : Fin 3)
    else if bids (0 : Fin 3) < bids (2 : Fin 3) then
      (0 : Fin 3)
    else
      (2 : Fin 3)
  else
    if bids (0 : Fin 3) < bids (2 : Fin 3) then
      (1 : Fin 3)
    else if bids (1 : Fin 3) < bids (2 : Fin 3) then
      (1 : Fin 3)
    else
      (2 : Fin 3)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.topBidder3`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(bids : Fin 3 → ℝ) → if bids 0 < bids 1 then if bids 1 < bids 2 then 2 else 1 else if bids 0 < bids 2 then 2 else 0
```

Exact Lean library declaration

```
noncomputable def topBidder3 (bids : Fin 3 → ℝ) : Fin 3 :=
  if bids (0 : Fin 3) < bids (1 : Fin 3) then
    if bids (1 : Fin 3) < bids (2 : Fin 3) then
      (2 : Fin 3)
    else
      (1 : Fin 3)
  else
    if bids (0 : Fin 3) < bids (2 : Fin 3) then
      (2 : Fin 3)
    else
      (0 : Fin 3)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.gsp3TwoSlotMechanism`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(a : Fin 3 → ℝ) →
  {
    slotOf := fun i =>
      if i = EconCSLib.Auction.topBidder3 a then some 0
      else if i = EconCSLib.Auction.secondBidder3 a then some 1 else none,
    paymentPerClick := fun i =>
      if i = EconCSLib.Auction.topBidder3 a then a (EconCSLib.Auction.secondBidder3 a)
      else if i = EconCSLib.Auction.secondBidder3 a then a (EconCSLib.Auction.thirdBidder3 a) else 0 }
```

Exact Lean library declaration

```
noncomputable def gsp3TwoSlotMechanism :
    PositionMechanism (Fin 3) (Fin 2) :=
  fun bids =>
    { slotOf := fun i =>
        if i = topBidder3 bids then some (0 : Fin 2)
        else if i = secondBidder3 bids then some (1 : Fin 2)
        else none
      paymentPerClick := fun i =>
        if i = topBidder3 bids then bids (secondBidder3 bids)
        else if i = secondBidder3 bids then bids (thirdBidder3 bids)
        else 0 }
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.gspCounterexampleEnvironment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ clickThroughRate := fun s => if s = 0 then 1 else 1 / 2 }
```

Exact Lean library declaration

```
noncomputable def gspCounterexampleEnvironment : PositionEnvironment (Fin 2) where
  clickThroughRate s := if s = (0 : Fin 2) then 1 else 1 / 2
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_eos_running_example_clickThroughRate`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(a : ℕ) →
  match a with
  | 0 => 200
  | 1 => 100
  | x => 0
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_eos_running_example_environment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ clickThroughRate := fun s => if s = 0 then 200 else 100 }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_eos_running_example_value`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(a : ℕ) →
  match a with
  | 0 => 10
  | 1 => 4
  | 2 => 2
  | x => 0
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_eos_running_example_values3

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(i : Fin 3) →
  match i with
  | ⟨0, isLt⟩ => 10
  | ⟨1, isLt⟩ => 4
  | ⟨val, isLt⟩ => 2
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_position_no_positive_transfers

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (i : Bidder),
  0 ≤ O.paymentPerClick i
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_ranked_gsp_rank`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} → (bids : Fin m → ℝ) → (i : Fin m) → {j | bids i < bids j}.card
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_ranked_gsp_bid_at_rank`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
  (bids : Fin m → ℝ) →
    (rank : ℕ) → if h : ∃ i, EconCSLib.Auction.paper_ranked_gsp_rank bids i = rank then bids (Classical.choose h) else 0
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_ranked_gsp_mechanism`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(m n : ℕ) →
  (a : Fin m → ℝ) →
    {
      slotOf := fun i =>
        if h : EconCSLib.Auction.paper_ranked_gsp_rank a i < n then
          some ⟨EconCSLib.Auction.paper_ranked_gsp_rank a i, h⟩
        else none,
      paymentPerClick := fun i =>
        EconCSLib.Auction.paper_ranked_gsp_bid_at_rank a (EconCSLib.Auction.paper_ranked_gsp_rank a i + 1) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_ranked_gsp_tiebreak_key`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} → (bids : Fin m → ℝ) → (i : Fin m) → toLex (-bids i, i)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_ranked_gsp_tiebreak_rank

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
  (bids : Fin m → ℝ) →
    (i : Fin m) →
      {j |
          EconCSLib.Auction.paper_ranked_gsp_tiebreak_key bids j <
            EconCSLib.Auction.paper_ranked_gsp_tiebreak_key bids i}.card
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_ranked_gsp_tiebreak_bid_at_rank

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{m : ℕ} →
  (bids : Fin m → ℝ) →
    (rank : ℕ) →
      if h : ∃ i, EconCSLib.Auction.paper_ranked_gsp_tiebreak_rank bids i = rank then bids (Classical.choose h) else 0
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_ranked_gsp_tiebreak_mechanism

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(m n : ℕ) →
  (a : Fin m → ℝ) →
    {
      slotOf := fun i =>
        if h : EconCSLib.Auction.paper_ranked_gsp_tiebreak_rank a i < n then
          some ⟨EconCSLib.Auction.paper_ranked_gsp_tiebreak_rank a i, h⟩
        else none,
      paymentPerClick := fun i =>
        EconCSLib.Auction.paper_ranked_gsp_tiebreak_bid_at_rank a
          (EconCSLib.Auction.paper_ranked_gsp_tiebreak_rank a i + 1) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →
  (value vcgTotalPayment clickThroughRate : ℕ → ℝ) →
    { slotOf := fun i => some i,
      paymentPerClick := fun i =>
        EconCSLib.Auction.paper_theorem7_bstar_bid value vcgTotalPayment clickThroughRate (↑i + 1) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem7_ranked_canonical_tail_remaining`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (i : Fin n) → n - (↑i + 1)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem7_ranked_environment`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (clickThroughRate : ℕ → ℝ) → { clickThroughRate := fun i => clickThroughRate ↑i }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment._f`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(value clickThroughRate : ℕ → ℝ) →
  (x : ℕ) →
    (f : Nat.below (motive := fun x => ℕ → ℝ) x) →
      (x_1 : ℕ) →
        (match (motive := ℕ → (x : ℕ) → Nat.below (motive := fun x => ℕ → ℝ) x → ℝ) x_1, x with
          | _i, 0 => fun x => 0
          | i, remaining.succ => fun x => (clickThroughRate i - clickThroughRate (i + 1)) * value (i + 1) + x.1 (i + 1))
          f
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_continue_payoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(clickThroughRate value : ℕ → ℝ) →
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
    (rank : ℕ) → clickThroughRate rank * (value (rank + 1) - state.clockPrice)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_at_threshold_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining : ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ)
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) (rank : ℕ),
  state.clockPrice =
      EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate (remaining + 1) (rank + 1) →
    strategy state rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_payoff

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(clickThroughRate value : ℕ → ℝ) →
  (remaining rank : ℕ) →
    clickThroughRate (rank + 1) *
      (value (rank + 1) -
        EconCSLib.Auction.paper_theorem8_bstar_threshold_bid value clickThroughRate remaining (rank + 2))
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_price`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (rank : ℕ) →
    EconCSLib.Auction.paper_theorem8_bstar_threshold_bid model.value model.clickThroughRate (model.remaining + 1)
      (rank + 1)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_next_state

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
    (rank : ℕ) →
      (state.advanceClock
            (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_price model
              rank)).recordDropout
        rank
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening   Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_final_state

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (a : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
    (a_1 : List ℕ) →
      List.brecOn (motive := fun x =>
        EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
          EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
        a_1
        (fun x f x_1 =>
          (match (motive :=
              EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
                (x : List ℕ) →
                  List.below (motive := fun x =>
                      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
                        EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
                      x →
                    EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
              x_1, x with
            | state, [] => fun x => state
            | state, rank :: tail => fun x =>
              x.1
                (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_next_state model state
                  rank))
            f)
        a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_final_state._f

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (x : List ℕ) →
    (f :
        List.below (motive := fun x =>
          EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
            EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
          x) →
      (x_1 : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
        (match (motive :=
            EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
              (x : List ℕ) →
                List.below (motive := fun x =>
                    EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ →
                      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
                    x →
                  EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
            x_1, x with
          | state, [] => fun x => state
          | state, rank :: tail => fun x =>
            x.1
              (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_next_state model state rank))
          f
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_fin_schedule_ranks`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} → (scheduledRanks : List (Fin n)) → List.map (fun rank => ↑rank) scheduledRanks
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_finite_active_exact_record_cold_start_state

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (activeRanks : Finset ℕ) →
    { clockPrice := -1,
      lastDropout := fun rank =>
        if rank ∈ activeRanks then none
        else
          some
            (EconCSLib.Auction.paper_theorem8_bstar_threshold_bid model.value model.clickThroughRate
              (model.remaining + 1) (rank + 1)) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_one_step_best_response_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining : ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ)
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) (rank : ℕ),
  (strategy state rank →
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_continue_payoff clickThroughRate value state rank ≤
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_payoff clickThroughRate value remaining rank) ∧
    (¬strategy state rank →
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_payoff clickThroughRate value remaining rank ≤
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_continue_payoff clickThroughRate value state rank)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_deviation_sequential_rationality_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining : ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ),
  EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_one_step_best_response_statement clickThroughRate value
      remaining strategy ∧
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_at_threshold_statement clickThroughRate value remaining
      strategy
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_optimality_certificate_of_strict`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) →
  { clickThroughRate := model.clickThroughRate, value := model.value, remaining := model.remaining, click_pos := ⋯,
    current_nonneg := ⋯, current_le := ⋯, continuation_bstar_le_value := ⋯ }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_one_step_best_response_at_state

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining : ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ)
  (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) (rank : ℕ),
  (strategy state rank →
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_continue_payoff clickThroughRate value state rank ≤
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_payoff clickThroughRate value remaining rank) ∧
    (¬strategy state rank →
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_payoff clickThroughRate value remaining rank ≤
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_continue_payoff clickThroughRate value state rank)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening   Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_deviation_dynamic_game`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) →
  { environment := { clickThroughRate := model.clickThroughRate }, values := model.value,
    isConsistentBelief := fun x x_1 => True,
    isSequentiallyRational := fun strategy x =>
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_deviation_sequential_rationality_statement
        model.clickThroughRate model.value model.remaining strategy,
    outcomeOf := fun x =>
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome model.value model.clickThroughRate
        model.remaining,
    vcgOutcome :=
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome model.value model.clickThroughRate
        model.remaining }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_named_strategy_local_deviation_sequential_rationality

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ (model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate),
  EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_deviation_sequential_rationality_statement
    model.clickThroughRate model.value model.remaining
    (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy model.value model.clickThroughRate
      model.remaining)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_deviation_constructed_outcome_certificate_of_local_optimality

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictLocalOptimalityCertificate) →
  { game := EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_deviation_dynamic_game model,
    strictModel := model, game_click_eq := ⋯, game_value_eq := ⋯, belief := (), equilibrium_consistent := trivial,
    local_optimality_implies_sequential_rationality := ⋯, pbe_bstar_threshold_characterization := ⋯,
    named_strategy_outcome_eq_bstar := ⋯, vcg_outcome_eq_bstar := ⋯ }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_optimality_certificate_of_strict_ordered

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) →
  { clickThroughRate := model.clickThroughRate, value := model.value, remaining := model.remaining, click_pos := ⋯,
    current_lt := ⋯, continuation_tail_payment_lt := ⋯ }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_constructed_outcome_certificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) →
  {
    base :=
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_deviation_constructed_outcome_certificate_of_local_opt
      ↪  imality
        (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_local_optimality_certificate_of_strict_ordered
          model),
    value_nonneg := ⋯, value_mono := ⋯ }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_exact_schedule_model

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) →
  EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_optimality_certificate_of_strict
    (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_constructed_outcome_certificate
          model).base.strictModel
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_price_sorted_fin_schedule

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedLocalOptimalityCertificate) →
  (n : ℕ) →
    have localModel :=
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_exact_schedule_model model;
    List.insertionSort
      (fun earlier later =>
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_price localModel ↑earlier ≤
          EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_price localModel ↑later)
      (List.finRange n)
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_source_sequential_rationality_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (clickThroughRate value : ℕ → ℝ) (remaining : ℕ)
  (initialState : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ),
  (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyHistory strategy initialState state →
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_one_step_best_response_at_state clickThroughRate value
          remaining strategy state) ∧
    (∀ (state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ),
        ¬EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyHistory strategy initialState state →
          EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_one_step_best_response_at_state clickThroughRate value
            remaining strategy state) ∧
      EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_drop_at_threshold_statement clickThroughRate value
        remaining strategy
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_source_extensive_rationality_statement

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ (model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate)
  (initialState finalState : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ)
  (strategy : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishStrategy ℕ),
  EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_source_sequential_rationality_statement model.clickThroughRate
      model.value model.remaining initialState strategy ∧
    EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyHistory strategy initialState finalState ∧
      EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState.StrategyTerminal strategy finalState
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_optimality_certificate_of_strict_values

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdStrictOrderedValueCertificate) →
  { clickThroughRate := model.clickThroughRate, value := model.value, remaining := model.remaining, click_pos := ⋯,
    current_lt := ⋯, value_nonneg := ⋯, value_mono := ⋯, continuation_tail_payment_lt := ⋯ }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_terminal_dropout_record_outcome

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
  { slotOf := fun i => some i,
    paymentPerClick := fun i =>
      match state.lastDropout i with
      | some price => price
      | none => 0 }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_terminal_record_source_extensive_dynamic_game_of_states

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(model : EconCSLib.Auction.PaperTheorem8BStarRankedThresholdLocalOptimalityCertificate) →
  (initialState finalState : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ) →
    { environment := { clickThroughRate := model.clickThroughRate }, values := model.value,
      isConsistentBelief := fun x x_1 => True,
      isSequentiallyRational := fun strategy x =>
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_source_extensive_rationality_statement model
          initialState finalState strategy,
      outcomeOf := fun x => EconCSLib.Auction.paper_theorem8_terminal_dropout_record_outcome finalState,
      vcgOutcome :=
        EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_outcome model.value model.clickThroughRate
          (model.remaining + 1) }
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 443-464

Verbatim source input

```
Definition 4 An equilibrium of the static game induced by GSP is locally envy-free if a player
cannot improve his payoff by exchanging bids with the player ranked one position above him. More
formally, in a locally envy-free equilibrium, for any i ≤ min{N, K}, we have αi sg(i) − p(i) ≥
αi−1 sg(i) − p(i−1) .
Of course, it is possible that in the dynamic game bids change over time, depending on the
player's strategies and information structure. However, as long as the restrictions are satisfied, if
the dynamic game ever converges to a static vector of bids, that static equilibrium should correspond
to a "locally envy-free" equilibrium of the static game Γ induced by the GSP. Consequently, we
view a locally envy-free equilibrium Γ as a prediction regarding a rest point at which the vector of
bids stabilizes. In this section, we study the set of locally envy-free equilibria.
We first show that the set of locally envy-free equilibria maps naturally to a set of stable
assignments in a corresponding two-sided matching market. The idea that auctions and two-sided
matching models are closely related is not new: it goes back to Crawford and Knoer (1981), Kelso
and Crawford (1982), Leonard (1983), and Demange, Gale, and Sotomayor (1986), and has been
studied in detail in a recent paper by Hatfield and Milgrom (2005). Note, however, that in our case
the non-standard auction is very different from those in the above papers.
Our environment maps naturally into the most basic assignment model, studied first by Shapley
and Shubik (1972). We view each position as an agent who is looking for a match with an advertiser.
The value of an advertiser-position pair (i, k) is equal to αk si . We call this assignment game A.
The advertiser makes its payment pik for the position, and the advertiser is left with αk si − pik .
The following pair of lemmas shows that there is a natural mapping from the set of locally envy-free
equilibria of GSP to the set of stable assignments. All proofs are in the Appendix.
```

Expanded PaperInterface specification

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : DecidableEq Bidder] (E : EconCSLib.Auction.PositionEnvironment Slot)
  (M : EconCSLib.Auction.PositionMechanism Bidder Slot) (values bids : Bidder → ℝ) (allocatedPositions : ℕ)
  (bidderAtRank : ℕ → Bidder) (slotAtRank : ℕ → Slot),
  EOS07GSP.PaperInterface.sourceDefinition4LocallyEnvyFree E M values bids allocatedPositions bidderAtRank slotAtRank ↔
    EconCSLib.Auction.PositionMechanism.IsNashEquilibrium E M values bids ∧
      (∀ rank < allocatedPositions, (M bids).slotOf (bidderAtRank rank) = some (slotAtRank rank)) ∧
        ∀ (rank : ℕ),
          rank + 1 < allocatedPositions →
            E.clickThroughRate (slotAtRank rank) *
                (values (bidderAtRank (rank + 1)) - (M bids).paymentPerClick (bidderAtRank rank)) ≤
              E.clickThroughRate (slotAtRank (rank + 1)) *
                (values (bidderAtRank (rank + 1)) - (M bids).paymentPerClick (bidderAtRank (rank + 1)))
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.definition4_locally_envy_free
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 459-462

Verbatim source input

```
Our environment maps naturally into the most basic assignment model, studied first by Shapley
and Shubik (1972). We view each position as an agent who is looking for a match with an advertiser.
The value of an advertiser-position pair (i, k) is equal to αk si . We call this assignment game A.
The advertiser makes its payment pik for the position, and the advertiser is left with αk si − pik .
```

Expanded PaperInterface specification

```
∀ {Bidder : Type u_1} {Slot : Type u_2} (E : EconCSLib.Auction.PositionEnvironment Slot)
  (O : EconCSLib.Auction.PositionOutcome Bidder Slot) (values : Bidder → ℝ),
  EconCSLib.Auction.PositionOutcome.StableAssignment E O values ↔
    O.FeasibleAssignment ∧
      EconCSLib.Auction.PositionOutcome.IndividuallyRational E O values ∧
        ∀ (i j : Bidder) (s : Slot),
          O.slotOf j = some s →
            E.clickThroughRate s * (values i - O.paymentPerClick j) ≤
              EconCSLib.Auction.PositionOutcome.utility E O values i
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.stable_assignment
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 3

Source locator: cited publication, lines 239-248

Verbatim source input

```
Example. Suppose there are two slots on a page and three bidders. An ad in the first slot
receives 200 clicks per hour, while the second slot gets 100. Bidders 1, 2, and 3 have values per
click of $10, $4, and $2, respectively. Suppose bidder 2 bids $2.01, to guarantee that he gets a slot.
Then bidder 1 will not want to bid more than $2.02—he does not need to pay more than that to
get the top spot. But then bidder 2 will want to revise his bid to $2.03 to get the top spot, bidder
1 will in turn raise his bid to $2.04, and so on. Clearly, there is no pure strategy equilibrium in the
one-shot version of the game, and so if bidders best respond to each other, they will want to revise
their bids as often as possible. Figure 1 shows an example of this behavior on Overture.
2.2.3
```

Expanded PaperInterface specification

```
200 * (10 - 203 / 100) < 200 * (10 - 202 / 100) ∧
  100 * (4 - 201 / 100) < 200 * (4 - 203 / 100) ∧ 100 * (10 - 202 / 100) < 200 * (10 - 204 / 100)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.first_price_running_example_profitable_revision_chain
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 372-386

Verbatim source input

```
Remark 1 If all advertisers were to bid the same amounts under the two mechanisms, then each
advertiser's payment would be at least as large under GSP as under VCG.
10
In actual sponsored search auctions at Google and Yahoo!, advertisers can also choose to place “broad match”
bids that match searches that include a keyword along with additional search terms.
11
The actual practice at Overture is to show equal bids according to the order in which the bidders placed their
bids.
12
Although we normalize the reserve price to zero, search engines charge the last bidder a non-zero reserve price.

8


[form-feed in source extraction]
This is easy to show by induction on advertisers' payments, starting with the last advertiser
who gets assigned a position. For i = min{K, N }, p(i) = pV,(i) = αi b(i+1) . For any i < min{K, N },
```

Expanded PaperInterface specification

```
∀ {value clickThroughRate : ℕ → ℝ},
  (∀ (i : ℕ), 0 ≤ value i) →
    (∀ (i : ℕ), value (i + 1) ≤ value i) →
      (∀ (i : ℕ), 0 ≤ clickThroughRate i) →
        ∀ {rank remaining : ℕ},
          0 < clickThroughRate rank →
            EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate rank remaining /
                clickThroughRate rank ≤
              value (rank + 1)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.remark1_gsp_payments_weakly_dominate_vcg
```

Recorded source-to-Spec screening   Verdict: `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 388-389

Verbatim source input

```
Remark 2 Truth-telling is a dominant strategy under VCG.
This is a well-known property of the VCG mechanism.
```

Expanded PaperInterface specification

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : Fintype Bidder] [inst_1 : DecidableEq Bidder] [inst_2 : Fintype Slot]
  [inst_3 : DecidableEq Slot] {E : EconCSLib.Auction.PositionEnvironment Slot},
  (∀ (s : Slot), 0 < E.clickThroughRate s) →
    EconCSLib.Auction.PositionMechanism.TruthfulDominantStrategy E
      (EconCSLib.Auction.PositionMechanism.positionVCGMechanism E)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.remark2_vcg_truthful
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 390-401

Verbatim source input

```
Remark 3 Truth-telling is not a dominant strategy under GSP.
For instance, consider a slight modification of the example from Section 2. There are still
three bidders, with values per click of $10, $4, and $2, and two positions. However, the clickthrough rates of these positions
↪ are now almost the same: the first position receives 200 clicks per
hour, and the second one gets 199. If all players bid truthfully, then bidder 1's payoff is equal to
($10 − $4) ∗ 200 = $1200. If, instead, he shades his bid and bids only $3 per click, he will get the
second position, and his payoff will be equal to ($10 − $2) ∗ 199 = $1592 > $1200.

4

Static GSP and Locally Envy-Free Equilibria13

Advertisers bidding on Yahoo! and Google can change their bids very frequently. We therefore
```

Expanded PaperInterface specification

```
¬EconCSLib.Auction.PositionMechanism.TruthfulDominantStrategy EconCSLib.Auction.gspCounterexampleEnvironment
    EconCSLib.Auction.gsp3TwoSlotMechanism
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.remark3_gsp_not_truthful
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 239-278

Verbatim source input

```
Example. Suppose there are two slots on a page and three bidders. An ad in the first slot
receives 200 clicks per hour, while the second slot gets 100. Bidders 1, 2, and 3 have values per
click of $10, $4, and $2, respectively. Suppose bidder 2 bids $2.01, to guarantee that he gets a slot.
Then bidder 1 will not want to bid more than $2.02—he does not need to pay more than that to
get the top spot. But then bidder 2 will want to revise his bid to $2.03 to get the top spot, bidder
1 will in turn raise his bid to $2.04, and so on. Clearly, there is no pure strategy equilibrium in the
one-shot version of the game, and so if bidders best respond to each other, they will want to revise
their bids as often as possible. Figure 1 shows an example of this behavior on Overture.
2.2.3

Generalized Second-Price Auctions

Under the generalized first-price auction, the bidder who could react to its competitors' moves
fastest had a substantial advantage. The mechanism therefore encouraged inefficient investments
in gaming the system. It also created volatile prices that in turn caused allocative inefficiencies.
Google addressed these problems when it introduced its own pay-per-click system, AdWords Select,
in February 2002. Google also recognized that a bidder in position i will never want to pay more
than one bid increment above the bid of the advertiser in position (i + 1), and Google adopted
this principle in its newly-designed generalized second price auction mechanism. In the simplest
GSP auction, an advertiser in position i pays a price per click equal to the bid of an advertiser in
position (i + 1) plus a minimum increment (typically $0.01). This second-price structure makes the
market more user friendly and less susceptible to gaming.
Recognizing these advantages, Yahoo!/Overture also switched to GSP. Let us describe the
version of GSP that it implemented.5 Every advertiser submits a bid. Advertisers are arranged
on the page in descending order of their bids. The advertiser in the first position pays a price per
click that equals the bid of the second advertiser plus an increment; the second advertiser pays the
price offered by the third advertiser plus an increment; and so forth.6
Example (continued). Let us now consider the payments in the environment of the previous
example under GSP mechanism. If all advertisers bid truthfully, then bids are $10, $4, $2. Payments
in GSP will be $4 and $2.7 Truth-telling is indeed an equilibrium in this example, because no bidder
can benefit by changing his bid. Note that total payments of bidders one and two are $800 and
$200, respectively.
5

We focus on Overture's implementation, because Google's system is somewhat more complex. However, it is
straightforward to generalize our analysis to Google's mechanism.
6
For this version of GSP to be efficient, it is necessary that the number of clicks received by an advertisement in
a given position depends on the ad's position and not on the advertiser's identity. Recognizing that this assumption
may not hold if some ads attract more clicks than others, Google adjusts effective bids based on ads' click-through
```

Expanded PaperInterface specification

```
EconCSLib.Auction.PositionMechanism.IsNashEquilibrium EconCSLib.Auction.paper_eos_running_example_environment
  EconCSLib.Auction.gsp3TwoSlotMechanism EconCSLib.Auction.paper_eos_running_example_values3
  EconCSLib.Auction.paper_eos_running_example_values3
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.running_example_truthful_gsp_nash
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 266-315

Verbatim source input

```
Example (continued). Let us now consider the payments in the environment of the previous
example under GSP mechanism. If all advertisers bid truthfully, then bids are $10, $4, $2. Payments
in GSP will be $4 and $2.7 Truth-telling is indeed an equilibrium in this example, because no bidder
can benefit by changing his bid. Note that total payments of bidders one and two are $800 and
$200, respectively.
5

We focus on Overture's implementation, because Google's system is somewhat more complex. However, it is
straightforward to generalize our analysis to Google's mechanism.
6
For this version of GSP to be efficient, it is necessary that the number of clicks received by an advertisement in
a given position depends on the ad's position and not on the advertiser's identity. Recognizing that this assumption
may not hold if some ads attract more clicks than others, Google adjusts effective bids based on ads' click-through
rates. But under the assumption that all ads have the same click-through rates conditional on position, Google's
and Yahoo!'s versions of GSP are identical. With modification for Google's adjustment procedure, our results for the
Yahoo! version of GSP also hold for Google adjusted GSP.
7
For convenience, we neglect the $0.01 minimum increments.

6


[form-feed in source extraction]
2.3

Generalized Second-Price and VCG Auctions

GSP looks similar to the Vickrey-Clarke-Groves mechanism, because both mechanisms set each
agent's payments only based on the allocation and bids of other players, not based on that agent's
own bid. In fact, Google's advertising materials explicitly refer to Vickrey and state that Google's
"unique auction model uses Nobel Prize-winning economic theory to eliminate ... that feeling that
you've paid too much".8 But GSP is not VCG. In particular, unlike the VCG auction, GSP does
not have an equilibrium in dominant strategies, and truth-telling is generally not an equilibrium
strategy in GSP. (See the example in Remark 3.) With only one slot, VCG and GSP would be
identical. With several slots, the mechanisms are different: while GSP essentially charges bidder i
the bid of bidder i + 1, VCG charges bidder i the externality that he imposes on others, i.e., the
decrease in the value of clicks received by other bidders because of i's presence.
Example (continued). Let us compute VCG payments for the example considered above.
The second bidder's payment is $200, as before. However, the payment of the first advertiser is now
$600: $200 for the externality that he imposes on bidder 3 (by forcing him out of position 2) and
$400 for the externality that he imposes on bidder 2 (by moving him from position 1 to position
2 and thus causing him to lose (200 − 100) = 100 clicks per hour). Note that in this example,
revenues under VCG are lower than under GSP. As we will show later, if advertisers were to bid
their true values under both mechanisms, revenues would always be higher under GSP.

2.4

Assessing the Market's Development

The chronology above suggests three major stages in the development of the sponsored search
advertising market. First, ads were sold manually, slowly, in large batches, and on a cost-perimpression basis. Second, Overture
↪  implemented keyword-targeted per-click sales and began to
streamline advertisement sales with some self-serve bidding interfaces, but with a highly unstable
```

Expanded PaperInterface specification

```
EconCSLib.Auction.paper_eos_running_example_clickThroughRate 0 * EconCSLib.Auction.paper_eos_running_example_value 1 +
    EconCSLib.Auction.paper_eos_running_example_clickThroughRate 1 *
      EconCSLib.Auction.paper_eos_running_example_value 2 >
  EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment EconCSLib.Auction.paper_eos_running_example_value
      EconCSLib.Auction.paper_eos_running_example_clickThroughRate 0 2 +
    EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment EconCSLib.Auction.paper_eos_running_example_value
      EconCSLib.Auction.paper_eos_running_example_clickThroughRate 1 1
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.running_example_truthful_gsp_revenue_comparison
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 463-466

Verbatim source input

```
The following pair of lemmas shows that there is a natural mapping from the set of locally envy-free
equilibria of GSP to the set of stable assignments. All proofs are in the Appendix.
Lemma 5 The outcome of any locally envy-free equilibrium of auction Γ is a stable assignment.
Lemma 6 If the number of bidders is greater than the number of available positions, then any

[Next verbatim source excerpt]

Proof of Lemma 5
By definition, in any locally envy-free equilibrium outcome, no player can profitably re-match with
the position assigned to the player right above him. Also, no player (a) can profitably re-match
with a position assigned to a player below him (b)—if such a profitable re-matching existed, player
a would find it profitable to slightly undercut player b in game Γ and get b’s position and payment.
But this would contradict the assumption that we are in equilibrium.22
Hence, we only need to show that no player can profitably re-match with the position assigned
to a player more than one spot above him. First, note that in any locally envy-free equilibrium,
the resulting matching must be assortative, i.e., for any i, the player assigned to position i has a
higher per-click valuation than the player assigned to position i + 1, and therefore, player 1 must
be assigned to the top position, player 2—to the second-highest position, and so on.
Indeed, let si and si+1 denote the values of players assigned to positions i and i+1. Equilibrium
restrictions imply that αi si − pi ≥ αi+1 si − pi+1 (nobody wants to move one position down),
and local envy-freeness implies that αi+1 si+1 − pi+1 ≥ αi si+1 − pi (nobody wants to move one
position up). Manipulating the above inequalities yields αi si − αi si+1 + αi+1 si+1 ≥ αi+1 si , thus
(αi − αi+1 )si ≥ (αi − αi+1 )si+1 . Since αi > αi+1 , we have si ≥ si+1 , and hence the locally envy-free
equilibrium outcome must be an assortative match.
Now, let us show that no player can profitably re-match with the position assigned to a player
more than one spot above him. Suppose bidder k is considering re-matching with position m < k−1.
Since the equilibrium is locally envy-free, we have
αk sk − pk ≥ αk−1 sk − pk−1
αk−1 sk−1 − pk−1 ≥ αk−2 sk−1 − pk−2
..
.
αm+1 sm+1 − pm+1 ≥ αm sm+1 − pm .
Since αj > αj+1 for any j, and sj > sk for any j < k, the above inequalities remain valid after
replacing sj with sk . Doing that, then adding all inequalities up, and canceling out the redundant
elements, we get αk sk − pk ≥ αm sk − pm . But that implies that advertiser k cannot re-match
profitably with position m, and we are done.
```

Expanded PaperInterface specification

```
∀ {Bidder : Type u_1} {Slot : Type u_2} [inst : DecidableEq Bidder] (E : EconCSLib.Auction.PositionEnvironment Slot)
  (M : EconCSLib.Auction.PositionMechanism Bidder Slot) (values bids : Bidder → ℝ),
  (M bids).FeasibleAssignment →
    EconCSLib.Auction.PositionOutcome.IndividuallyRational E (M bids) values →
      EconCSLib.Auction.PositionMechanism.LocallyEnvyFreeEquilibrium E M values bids →
        EconCSLib.Auction.PositionOutcome.StableAssignment E (M bids) values
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.lemma5_locally_envy_free_stable
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 466-480

Verbatim source input

```
Lemma 6 If the number of bidders is greater than the number of available positions, then any
stable assignment is an outcome of a locally envy-free equilibrium of auction Γ.
10


[form-feed in source extraction]
We will now construct a particular locally envy-free equilibrium of game Γ. This equilibrium
has two important properties. First, in this equilibrium bidders’ payments coincide with their
payments in the dominant- strategy equilibrium of VCG. Second, this equilibrium is the worst
locally envy-free equilibrium for the search engine, and it is the best locally envy-free equilibrium
for the bidders. Consequently, the revenues of a search engine are (weakly) higher in any locally
envy-free equilibrium of GSP than in the dominant-strategy equilibrium of VCG.
Consider the following strategy profile B ∗ . For each advertiser i ∈ {2, . . . , min{N, K}}, bid b∗i
V,(i−1)

is equal to p αi−1 , where pV,(j) is the payment of bidder j in the dominant-strategy equilibrium of
VCG. Bid b∗1 is equal to s1 .16

[Next verbatim source excerpt]

Proof of Lemma 6
Take a stable assignment. By a result of Shapley and Shubik (1972), this assignment must be
efficient, and hence assortative, and so we can talk about advertiser i being matched with position
22

This argument relies on the fact that in equilibrium, no two (or more) players bid the same amount, which is
straightforward to prove: since all players’ per-click values are different, and all ties are broken randomly with equal
probabilities, at least one of such players would find it profitable to bid slightly higher or slightly lower.

15


[form-feed in source extraction]
i, with associated payment pi .
Let us construct a locally envy-free equilibrium with the corresponding outcome. Let b1 = s1
and bi = αpi−1
for i > 1. Let us show that this set of strategies is a locally envy-free equilibrium.
i−1

First, note that for any i, bi > bi+1 (because otherwise we would have, for some i, αpi−1
≤ αpii ⇒
i−1

si − αpi−1
≥ si − αpii ⇒ αi−1 (si − pi−1 ) > αi (si − pi ), which would imply that player i could rei−1
match profitably). Therefore, position allocations and payments resulting from this strategy profile
will coincide with those in the original stable assignment. To see that this strategy profile is an
equilibrium, note that deviating and moving to a different position in this strategy profile is at
most as profitable for any player as re-matching with the corresponding position in the assignment
game. To see that this equilibrium is locally envy-free, note that the payoff from swapping with
the bidder above is exactly equal to the payoff from re-matching with that player’s position in the
assignment game.
```

Expanded PaperInterface specification

```
∀ {m n : ℕ},
  n < m →
    ∀ {value clickThroughRate : ℕ → ℝ} (O : EconCSLib.Auction.PositionOutcome (Fin m) (Fin n)) (bids : Fin m → ℝ),
      EconCSLib.Auction.paper_ranked_gsp_tiebreak_mechanism m n bids = O →
        (∀ {i j : Fin m}, ↑i < ↑j → bids j < bids i) →
          (∀ (s : Fin n), 0 ≤ clickThroughRate ↑s) →
            (EconCSLib.Auction.PositionOutcome.StableAssignment
                (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O fun i => value ↑i) →
              EconCSLib.Auction.PositionMechanism.LocallyEnvyFreeEquilibrium
                (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate)
                (EconCSLib.Auction.paper_ranked_gsp_tiebreak_mechanism m n) (fun i => value ↑i) bids
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.lemma6_tiebreak_ranked_gsp_stable_assignment_locally_envy_free
```

Recorded source-to-Spec screening   Verdict: `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 481-511

Verbatim source input

Theorem 7 Strategy profile B ∗ is a locally envy-free equilibrium of game Γ. In this equilibrium,
each bidder’s position and payment is equal to those in the dominant-strategy equilibrium of the
game induced by VCG. In any other locally envy-free equilibrium of game Γ, the total revenue of
the seller is at least as high as in B ∗ .
To prove Theorem 7, we first note that payments under strategy profile B ∗ coincide with VCG
payments and check that B ∗ is indeed a locally-envy free equilibrium. This follows from the fact
that, by construction, each bidder is indifferent between remaining in his positions and swapping
with the bidder one position above him. Next, from Lemma 5 we know that every locally envy-free
equilibrium corresponds to a stable assignment. Combining this with the fact17 that in any stable
assignment the payments of bidders must be at least as high as VCG payments completes the proof.

5

Main Result: GSP and the Generalized English Auction

In this section we introduce the generalized English auction that corresponds to the generalized
second price auction. The generalized English auction is a useful metaphor for predicting the
behavior of advertisers in GSP auctions in a dynamic environment. In the model analyzed in the
previous section, we assume that bidders have converged to a long-run steady state, have learned
each other’s values, and no longer have incentives to change their bids. But how do they converge
to such a situation?
In this section we show that, in fact, there are simple strategies that can quickly lead to such
an outcome: each advertiser starts bidding at zero and keeps increasing his bid as long as he finds
it profitable to do so. This simple myopic procedure leads to an equilibrium outcome in which
bidders have no incentives to change their bids. Remarkably, the outcome is the same as the worst
(for the search engine) locally-envy free equilibrium of the static GSP: all players get VCG payoffs.
We model this procedure through an analogue of the standard English auction. The generalized
English Auction entails a clock showing the current price (that continuously increases over time).
A player’s bid is the price on the clock at the time when he drops out. The auction is over when
This bid does not affect any bidder’s payment and can be set equal to any value greater than b∗2 .
Leonard (1983) states this fact for general assignment problems. For completeness, we include a short independent

[Next verbatim source excerpt]

Proof of Theorem 7
First, we need to check that the order of the bids is preserved, i.e., b∗i > b∗i+1 for any i < min{N, K}.
For i ≥ 2, this is equivalent to

pV,(i−1)
pV,(i)
>
αi−1
αi
m
(αi−1 − αi)si + pV,(i)
pV,(i)
>
αi−1
αi
m
αi (αi−1 − αi)si > (αi−1 − αi)pV,(i)
m
αi si > pV,(i) .

For i = 1, b∗i > b∗i+1 is equivalent to
s1 >

pV,(1)
α1

m
α1 s1 > pV,(1) .

To see that for any i, αi si > pV,(i) , note first that in the game induced by VCG, each player can
guarantee himself the payoff of at least zero (by bidding zero), and hence in any equilibrium his
16

[form-feed in source extraction]
payoff from clicks is at least as high as his payment. To prove that the inequality is strict, note
that if player i’s value per click were slightly lower, e.g., si − ∆ instead of si , ∆ < si − si+1 , then
his payment in the truth-telling equilibrium would still be the same (because it does not depend on
his own bid, given the allocation of positions), and so pV,(i) ≤ αi (si − ∆) < αi si . Thus, for any i,
b∗i > b∗i+1 , and therefore each bidder’s position is the same as in the truthful equilibrium of VCG.
Therefore, by construction, payments are also the same.
Next, to see that no bidder i can benefit by bidding less than b∗i , suppose that he bids an amount
b0 < b∗i that puts him in position i0 > i. Then, by construction, his payment will be equal to the
amount that he would need to pay to be in position i0 under VCG, provided that other players bid
truthfully. But truthful bidding is an equilibrium under VCG, and so such deviation cannot be
profitable there—hence, it cannot be profitable in strategy profile B ∗ of game Γ either.
To see that no bidder i can benefit by bidding more than b∗i , suppose that he bids an amount
b0 > b∗i that puts him in position i0 < i. Then the net payoff from this deviation is equal to
Pi−1 V,(j)
Pi−1
−
(αi0 −αi)si −(αi0 b∗i0 −αi bi+1) < (αi0 −αi)si −(αi0 b∗i0 +1 −αi bi+1) = j=i
0 (αj −αj+1)si −
j=i0 (p
Pi−1

Pi−1
V,(j+1)
p
) = j=i0 (αj − αj+1)si − j=i0 (αj − αj+1)sj+1 . But since si ≤ sj+1 for any j < i, the last expression is less than or equal to zero, and hence the deviation is not profitable.
To check that this equilibrium is locally envy-free, note that if bidder i swapped his bids with bidder i−1, his payoff would change by (αi−1 −αi)si −(αi−1 b∗i −αi bi+1) = (αi−1 −αi)si −(pV,(i−1) − pV,(i)) = (αi−1 − αi)si − (pV,(i−1) − pV,(i)) = (αi−1 − αi)si − (αi−1 − αi)si = 0. In other words, each bidder is indifferent between his actual payoff and his payoff after swapping bids with the bidder

Expanded PaperInterface specification

```
∀ {n : ℕ} {value vcgTotalPayment clickThroughRate : ℕ → ℝ},
  (∀ (i : ℕ), 0 < clickThroughRate i) →
    (∀ (i : ℕ), clickThroughRate (i + 1) < clickThroughRate i) →
      (∀ (i : ℕ),
          vcgTotalPayment i =
            (clickThroughRate i - clickThroughRate (i + 1)) * value (i + 1) + vcgTotalPayment (i + 1)) →
        (∀ (i : ℕ), vcgTotalPayment i < clickThroughRate i * value i) →
          (∀ (i : Fin n),
              (EconCSLib.Auction.paper_ranked_gsp_mechanism (n + 1) n fun bidder =>
                      EconCSLib.Auction.paper_theorem7_bstar_bid value vcgTotalPayment clickThroughRate ↑bidder).slotOf
                  i.castSucc =
                some i) ∧
            (EconCSLib.Auction.paper_ranked_gsp_mechanism (n + 1) n fun bidder =>
                      EconCSLib.Auction.paper_theorem7_bstar_bid value vcgTotalPayment clickThroughRate ↑bidder).slotOf
                  (Fin.last n) =
                none ∧
              ∀ (i : Fin n),
                (EconCSLib.Auction.paper_ranked_gsp_mechanism (n + 1) n fun bidder =>
                        EconCSLib.Auction.paper_theorem7_bstar_bid value vcgTotalPayment clickThroughRate
                          ↑bidder).paymentPerClick
                    i.castSucc =
                  (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                        clickThroughRate).paymentPerClick
                    i
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem7_ranked_gsp_bstar_mechanism_realizes_bstar_outcome
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 470-483

Verbatim source input

```
We will now construct a particular locally envy-free equilibrium of game Γ. This equilibrium
has two important properties. First, in this equilibrium bidders’ payments coincide with their
payments in the dominant- strategy equilibrium of VCG. Second, this equilibrium is the worst
locally envy-free equilibrium for the search engine, and it is the best locally envy-free equilibrium
for the bidders. Consequently, the revenues of a search engine are (weakly) higher in any locally
envy-free equilibrium of GSP than in the dominant-strategy equilibrium of VCG.
Consider the following strategy profile B ∗ . For each advertiser i ∈ {2, . . . , min{N, K}}, bid b∗i
V,(i−1)

is equal to p αi−1 , where pV,(j) is the payment of bidder j in the dominant-strategy equilibrium of
VCG. Bid b∗1 is equal to s1 .16
Theorem 7 Strategy profile B ∗ is a locally envy-free equilibrium of game Γ. In this equilibrium,
each bidder’s position and payment is equal to those in the dominant-strategy equilibrium of the
game induced by VCG. In any other locally envy-free equilibrium of game Γ, the total revenue of

[Next verbatim source excerpt]

Proof of Theorem 7
First, we need to check that the order of the bids is preserved, i.e., b∗i > b∗i+1 for any i < min{N, K}.
For i ≥ 2, this is equivalent to

pV,(i−1)
pV,(i)
>
αi−1
αi
m
(αi−1 − αi )si + pV,(i)
pV,(i)
>
αi−1
αi
m
αi (αi−1 − αi )si > (αi−1 − αi )pV,(i)
m
αi si > pV,(i) .

For i = 1, b∗i > b∗i+1 is equivalent to
s1 >

pV,(1)
α1

m
α1 s1 > pV,(1) .

To see that for any i, αi si > pV,(i) , note first that in the game induced by VCG, each player can
guarantee himself the payoff of at least zero (by bidding zero), and hence in any equilibrium his
16


[form-feed in source extraction]
payoff from clicks is at least as high as his payment. To prove that the inequality is strict, note
that if player i’s value per click were slightly lower, e.g., si − Δ instead of si , Δ < si − si+1 , then
his payment in the truth-telling equilibrium would still be the same (because it does not depend on
his own bid, given the allocation of positions), and so pV,(i) ≤ αi (si − Δ) < αi si . Thus, for any i,
b∗i > b∗i+1 , and therefore each bidder’s position is the same as in the truthful equilibrium of VCG.
Therefore, by construction, payments are also the same.
Next, to see that no bidder i can benefit by bidding less than b∗i , suppose that he bids an amount
b0 < b∗i that puts him in position i0 > i. Then, by construction, his payment will be equal to the
amount that he would need to pay to be in position i0 under VCG, provided that other players bid
truthfully. But truthful bidding is an equilibrium under VCG, and so such deviation cannot be
profitable there—hence, it cannot be profitable in strategy profile B ∗ of game Γ either.
To see that no bidder i can benefit by bidding more than b∗i , suppose that he bids an amount
b0 > b∗i that puts him in position i0 < i. Then the net payoff from this deviation is equal to
Pi−1 V,(j)
Pi−1
−
(αi0 −αi )si −(αi0 b∗i0 −αi bi+1 ) < (αi0 −αi )si −(αi0 b∗i0 +1 −αi bi+1 ) = j=i
0 (αj −αj+1 )si −
j=i0 (p
Pi−1
Pi−1
V,(j+1)
p
) = j=i0 (αj − αj+1 )si − j=i0 (αj − αj+1 )sj+1 . But since si ≤ sj+1 for any j < i, the last
expression is less than or equal to zero, and hence the deviation is not profitable.
To check that this equilibrium is locally envy-free, note that if bidder i swapped his bids with
bidder i−1, his payoff would change by (αi−1 −αi )si −(αi−1 b∗i −αi bi+1 ) = (αi−1 −αi )si −(pV,(i−1) −
pV,(i) ) = (αi−1 − αi )si − (pV,(i−1) − pV,(i) ) = (αi−1 − αi )si − (αi−1 − αi )si = 0. In other words, each
bidder is indifferent between his actual payoff and his payoff after swapping bids with the bidder
```

Expanded PaperInterface specification

```
∀ (value vcgTotalPayment clickThroughRate : ℕ → ℝ) (i : ℕ),
  clickThroughRate i ≠ 0 →
    clickThroughRate i * EconCSLib.Auction.paper_theorem7_bstar_bid value vcgTotalPayment clickThroughRate (i + 1) =
      vcgTotalPayment i
```

Lean proof endpoint (built separately)

`EOS07GSP.PaperInterface.theorem7_bstar_payment_identity`

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 481-488

Verbatim source input

```
Theorem 7 Strategy profile B ∗ is a locally envy-free equilibrium of game Γ. In this equilibrium,
each bidder’s position and payment is equal to those in the dominant-strategy equilibrium of the
game induced by VCG. In any other locally envy-free equilibrium of game Γ, the total revenue of
the seller is at least as high as in B ∗ .
To prove Theorem 7, we first note that payments under strategy profile B ∗ coincide with VCG
payments and check that B ∗ is indeed a locally-envy free equilibrium. This follows from the fact
that, by construction, each bidder is indifferent between remaining in his positions and swapping
with the bidder one position above him. Next, from Lemma 5 we know that every locally envy-free

[Next verbatim source excerpt]

To check that this equilibrium is locally envy-free, note that if bidder i swapped his bids with
bidder i−1, his payoff would change by (αi−1 −αi )si −(αi−1 b∗i −αi bi+1 ) = (αi−1 −αi )si −(pV,(i−1) −
pV,(i) ) = (αi−1 − αi )si − (pV,(i−1) − pV,(i) ) = (αi−1 − αi )si − (αi−1 − αi )si = 0. In other words, each
bidder is indifferent between his actual payoff and his payoff after swapping bids with the bidder
above, and hence the equilibrium is locally envy-free.
```

Expanded PaperInterface specification

```
∀ {n : ℕ} {value vcgTotalPayment clickThroughRate : ℕ → ℝ},
  (∀ (r : Fin n), clickThroughRate ↑r ≠ 0) →
    (∀ (k : ℕ), clickThroughRate (k + 1) ≤ clickThroughRate k) →
      (∀ (a b : ℕ), a ≤ b → value b ≤ value a) →
        (∀ (k : ℕ),
            vcgTotalPayment k =
              (clickThroughRate k - clickThroughRate (k + 1)) * value (k + 1) + vcgTotalPayment (k + 1)) →
          ∀ (i j s : Fin n),
            (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment clickThroughRate).slotOf j =
                some s →
              (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate).clickThroughRate s *
                  (value ↑i -
                    (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                          clickThroughRate).paymentPerClick
                      j) ≤
                EconCSLib.Auction.PositionOutcome.utility
                  (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate)
                  (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment clickThroughRate)
                  (fun i => value ↑i) i
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem7_bstar_locally_envy_free
```

Recorded source-to-Spec screening   **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 481-490

Verbatim source input

```
Theorem 7 Strategy profile B ∗ is a locally envy-free equilibrium of game Γ. In this equilibrium,
each bidder’s position and payment is equal to those in the dominant-strategy equilibrium of the
game induced by VCG. In any other locally envy-free equilibrium of game Γ, the total revenue of
the seller is at least as high as in B ∗ .
To prove Theorem 7, we first note that payments under strategy profile B ∗ coincide with VCG
payments and check that B ∗ is indeed a locally-envy free equilibrium. This follows from the fact
that, by construction, each bidder is indifferent between remaining in his positions and swapping
with the bidder one position above him. Next, from Lemma 5 we know that every locally envy-free
equilibrium corresponds to a stable assignment. Combining this with the fact17 that in any stable
assignment the payments of bidders must be at least as high as VCG payments completes the proof.

[Next verbatim source excerpt]

locally envy-free equilibrium for the search engine. A standard result of matching theory states
that there exists an assignment that is the best stable assignment for all advertisers, and the worst
stable assignment for all positions. A stable assignment is characterized by a vector of payments
p = (p1 , . . . , pK ). Let pV = (pV1 , . . . , pVK ) be the set of dominant-strategy VCG payments, i.e., the
set of payments in equilibrium B ∗ of game Γ.
In any stable assignment, pK must be at least as high as αK sK−1 , since otherwise advertiser
K − 1 would find it profitable to match with position K. On the other hand, pVK = αK sK−1 , and
hence in the buyer-optimal stable assignment, pK = pVK .
Next, in any stable assignment, it must be the case that pK−1 − pK ≥ (αK−1 − αK )sK −
otherwise, advertiser K would find it profitable to re-match with position K − 1. Hence, pK−1 ≥
(αK−1 −αK )sK +pK ≥ (αK−1 −αK )sK +pVK = pVK−1 , and so in the buyer-optimal stable assignment,
pK−1 = pVK−1 .
Proceeding by induction, we get pk = pVk for any k ≤ K in the buyer-optimal (and therefore
seller-pessimal) stable assignment, and so in any locally envy-free equilibrium of game Γ, the total
P
V
revenue of the seller is at least as high as K
```

Expanded PaperInterface specification

```
∀ {n : ℕ} (value vcgTotalPayment clickThroughRate : ℕ → ℝ),
  (∀ (s : Fin n), 0 ≤ clickThroughRate ↑s) →
    (∀ (s : Fin n), 0 < clickThroughRate ↑s) →
      (∀ (k : ℕ), clickThroughRate (k + 1) ≤ clickThroughRate k) →
        (∀ (a b : ℕ), a ≤ b → value b ≤ value a) →
          (∀ (k : ℕ),
              vcgTotalPayment k =
                (clickThroughRate k - clickThroughRate (k + 1)) * value (k + 1) + vcgTotalPayment (k + 1)) →
            (∀ (i : Fin n), vcgTotalPayment ↑i ≤ clickThroughRate ↑i * value ↑i) →
              (∀ (i : Fin n),
                  vcgTotalPayment ↑i =
                    EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate (↑i)
                      (EconCSLib.Auction.paper_theorem7_ranked_canonical_tail_remaining i)) →
                (∀ (i : ℕ), 0 ≤ value i) →
                  ∃ O,
                    EconCSLib.Auction.paper_position_no_positive_transfers O ∧
                      (EconCSLib.Auction.PositionOutcome.SlotEnvyFree
                          (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O fun i => value ↑i) ∧
                        (EconCSLib.Auction.PositionOutcome.StableAssignment
                            (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O fun i =>
                            value ↑i) ∧
                          (∀ (i : Fin n),
                              O.slotOf i =
                                (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                                      clickThroughRate).slotOf
                                  i) ∧
                            (∀ (i : Fin n),
                                O.paymentPerClick i =
                                  (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                                        clickThroughRate).paymentPerClick
                                    i) ∧
                              ∀ (other : EconCSLib.Auction.PositionOutcome (Fin n) (Fin n)),
                                other.FeasibleAssignment →
                                  (EconCSLib.Auction.PositionOutcome.IndividuallyRational
                                      (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) other
                                      fun i => value ↑i) →
                                    (EconCSLib.Auction.PositionOutcome.SlotEnvyFree
                                        (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) other
                                        fun i => value ↑i) →
                                      (∀ (i : Fin n), O.slotOf i = other.slotOf i) →
                                        EconCSLib.Auction.paper_position_no_positive_transfers other →
                                          EconCSLib.Auction.PositionOutcome.revenue
                                              (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O ≤
                                            EconCSLib.Auction.PositionOutcome.revenue
                                              (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate)
                                              other
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem7_no_positive_transfer_conclusion
```

Recorded source-to-Spec screening **Verdict:** `not recorded`
Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 15

Source locator: cited publication, lines 481-490

Verbatim source input

```
Theorem 7 Strategy profile B ∗ is a locally envy-free equilibrium of game Γ. In this equilibrium,
each bidder’s position and payment is equal to those in the dominant-strategy equilibrium of the
game induced by VCG. In any other locally envy-free equilibrium of game Γ, the total revenue of
the seller is at least as high as in B ∗ .
To prove Theorem 7, we first note that payments under strategy profile B ∗ coincide with VCG
payments and check that B ∗ is indeed a locally-envy free equilibrium. This follows from the fact
that, by construction, each bidder is indifferent between remaining in his positions and swapping
with the bidder one position above him. Next, from Lemma 5 we know that every locally envy-free
equilibrium corresponds to a stable assignment. Combining this with the fact17 that in any stable
assignment the payments of bidders must be at least as high as VCG payments completes the proof.

[Next verbatim source excerpt]

locally envy-free equilibrium for the search engine. A standard result of matching theory states
that there exists an assignment that is the best stable assignment for all advertisers, and the worst
stable assignment for all positions. A stable assignment is characterized by a vector of payments
p = (p1 , . . . , pK ). Let pV = (pV1 , . . . , pVK ) be the set of dominant-strategy VCG payments, i.e., the
set of payments in equilibrium B ∗ of game Γ.
In any stable assignment, pK must be at least as high as αK sK−1 , since otherwise advertiser
K − 1 would find it profitable to match with position K. On the other hand, pVK = αK sK−1 , and
hence in the buyer-optimal stable assignment, pK = pVK .
Next, in any stable assignment, it must be the case that pK−1 − pK ≥ (αK−1 − αK )sK −
otherwise, advertiser K would find it profitable to re-match with position K − 1. Hence, pK−1 ≥
(αK−1 −αK )sK +pK ≥ (αK−1 −αK )sK +pVK = pVK−1 , and so in the buyer-optimal stable assignment,
pK−1 = pVK−1 .
Proceeding by induction, we get pk = pVk for any k ≤ K in the buyer-optimal (and therefore
seller-pessimal) stable assignment, and so in any locally envy-free equilibrium of game Γ, the total
P
V
revenue of the seller is at least as high as K
```

Expanded PaperInterface specification

```
∀ {n : ℕ} (value vcgTotalPayment clickThroughRate : ℕ → ℝ),
  (∀ (s : Fin n), 0 ≤ clickThroughRate ↑s) →
    (∀ (s : Fin n), 0 < clickThroughRate ↑s) →
      (∀ (k : ℕ), clickThroughRate (k + 1) ≤ clickThroughRate k) →
        (∀ (a b : ℕ), a ≤ b → value b ≤ value a) →
          (∀ (k : ℕ),
              vcgTotalPayment k =
                (clickThroughRate k - clickThroughRate (k + 1)) * value (k + 1) + vcgTotalPayment (k + 1)) →
            (∀ (i : Fin n), vcgTotalPayment ↑i ≤ clickThroughRate ↑i * value ↑i) →
              (∀ (i : Fin n),
                  vcgTotalPayment ↑i =
                    EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate (↑i)
                      (EconCSLib.Auction.paper_theorem7_ranked_canonical_tail_remaining i)) →
                (∀ (i : ℕ), 0 ≤ value i) →
                  (∀ (k : ℕ), k + 1 < n → value (k + 1) < value k) →
                    (∀ (k : ℕ), k + 1 < n → clickThroughRate (k + 1) < clickThroughRate k) →
                      ∃ O,
                        EconCSLib.Auction.paper_position_no_positive_transfers O ∧
                          (EconCSLib.Auction.PositionOutcome.SlotEnvyFree
                              (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O fun i =>
                              value ↑i) ∧
                            (EconCSLib.Auction.PositionOutcome.StableAssignment
                                (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O fun i =>
                                value ↑i) ∧
                              (∀ (i : Fin n),
                                  O.slotOf i =
                                    (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                                          clickThroughRate).slotOf
                                      i) ∧
                                (∀ (i : Fin n),
                                    O.paymentPerClick i =
                                      (EconCSLib.Auction.paper_theorem7_ranked_bstar_outcome value vcgTotalPayment
                                            clickThroughRate).paymentPerClick
                                        i) ∧
                                  ∀ (bids : Fin n → ℝ),
                                    EconCSLib.Auction.PositionMechanism.LocallyEnvyFreeEquilibrium
                                        (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate)
                                        (EconCSLib.Auction.paper_ranked_gsp_tiebreak_mechanism n n) (fun i => value ↑i)
                                        bids →
                                      (∀ (i : Fin n), 0 ≤ bids i) →
                                        EconCSLib.Auction.PositionOutcome.revenue
                                            (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate) O ≤
                                          EconCSLib.Auction.PositionOutcome.revenue
                                            (EconCSLib.Auction.paper_theorem7_ranked_environment clickThroughRate)
                                            (EconCSLib.Auction.paper_ranked_gsp_tiebreak_mechanism n n bids)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem7_strict_tiebreak_gsp_comparison_conclusion
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 539-546

Verbatim source input

```
Theorem 8 In the unique perfect Bayesian equilibrium of the generalized English auction with
k
strategies continuous in si , an advertiser with value si drops out at price pi (k, h, si ) = si − ααk−1
(si −

bk+1 ). In this equilibrium, each advertiser’s resulting position and payoff are the same as in the
dominant-strategy equilibrium of the game induced by VCG. This equilibrium is ex post: the strategy
of each bidder is a best response to other bidders’ strategies regardless of their realized values.

[Next verbatim source excerpt]


[form-feed in source extraction]
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can “swap” such players’ strategies, and therefore there exists an “observationally equivalent”
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
(s − bk+1 ).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set
```

Expanded PaperInterface specification

```
∀ (value clickThroughRate : ℕ → ℝ) (remaining rank : ℕ),
  (∀ (i : ℕ), 0 < clickThroughRate i) →
    EOS07GSP.PaperInterface.theorem8RankedGeneralizedEnglishDropoutPrice clickThroughRate
        (fun k =>
          EOS07GSP.PaperInterface.theorem7BStarBid value
            (fun j => EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment value clickThroughRate j remaining)
            clickThroughRate (k + 2))
        value rank =
      EOS07GSP.PaperInterface.theorem8BStarThresholdBid value clickThroughRate (remaining + 1) (rank + 1)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_dropout_formula_eq_bstar_threshold
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 923-933

Verbatim source input

```
αl (s1 − bl+2 )−contradiction.

Step 2. In Step 1, we have shown that pmax (k, h, s) ≡ maxi pi (k, h, s) cannot be greater than
q(k, h, s), and therefore for any player i and type s ≥ smin , pi (k, h, s) ≤ q(k, h, s). Take some value
s > smin and player i. Suppose pi (k, h, s) < q(k, h, s). Take some other player j. From Step 1,
we have pj (k, h, smin ) ≤ q(k, h, smin ) < q(k, h, s), and therefore if player i waited until q(k, h, s)
instead of dropping out at pi (k, h, s), the probability that someone dropped out in between would
be positive, and hence the payoff would be strictly greater (by the definition of function q(), player
i with value s strictly prefers being in position k − 1 or higher at any price less than q(k, h, s) to
being in position k at price bk+1 ), which is impossible in equilibrium. Hence, pi (k, h, s) = q(k, h, s)
for all s > smin . By continuity, we also have pi (k, h, smin ) = q(k, h, smin ).
```

Expanded PaperInterface specification

```
∀ {clickThroughRate lastDropout value : ℕ → ℝ} {state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ}
  {rank : ℕ},
  0 < clickThroughRate rank →
    state.clockPrice <
        EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout value
          rank →
      clickThroughRate (rank + 1) * (value (rank + 1) - lastDropout rank) <
        clickThroughRate rank * (value (rank + 1) - state.clockPrice)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_q_step2_waiting_before_q_review
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 833-924

Verbatim source input

```
Without loss of generality, we can assume that there is a positive mass of types of other players
dropping out at or before pi (k, h, s0 ).23
Step 1(a). Suppose first that there is a positive mass of types of other players dropping out
at pi (k, h, s0 ) = pmax (k, h, s). That implies that with positive probability, player i of type s0 will
remain in the subgame following the drop-out of some other player at pi (k, h, s0 ) (since ties are
broken randomly). Let us show that in this subgame, player i of type s0 will be the first player to
23

Otherwise, we have ∀j 6= i, pj (k, h, s0 ) ≤ pi (k, h, s) ≤ pmax (k, h, s) = pi (k, h, s0 ) ⇒ ∀j 6= i, pj (k, h, s0 ) =
pi (k, h, s0 ) and ∀s0 > s0 , pj (k, h, s0 ) > pi (k, h, s0 ). But we also have pi (k, h, s0 ) = pmax (k, h, s) > q(k, h, s) ≥
q(k, h, s0 ), and so for some s0 > s0 we have pmax (k, h, s0 ) > q(k, h, s0 ) and pmax (k, h, s0 ) > pmax (k, h, s). We
can then consider s00 in place of s0 , where s00 is the smallest type, and i0 is the corresponding player, such that
pi0 (k, h, s00 ) = pmax (k, h, s0 ). There is a positive mass of types of other players dropping out before pi0 (k, h, s00 ).

18


[form-feed in source extraction]
drop out with probability 1. Suppose that is not the case, and let l < k − 1 be the smallest number
such that he gets position l with positive probability.
Consider any continuation of history h, hl+2 , such that the last player to drop out in that history
gets position l+2 and drops out at price bl+2 , player i of type s0 is one of the remaining l+1 players,
there is a positive probability that player i gets position l in the continuation subgame following
history hl+2 , and there is zero probability that player i gets position m for any m < l. Note
that s0 ≥ bl+2 —otherwise it would have been optimal for player i to drop out earlier. Consider
pi (l + 1, hl+2 , s0 ). There must be a positive mass of types of other players who drop out no
later than pi (l + 1, hl+2 , s0 ). Take the highest such type, s0 , and the corresponding player j.
It has to be the case that s0 > s0 ≥ bl+2 . It also has to be the case that q(l + 1, hl+2 , s0 ) is
less than or equal to pi (l + 1, hl+2 , s0 ). (Otherwise player j with value s0 would be playing a
strategy weakly dominated by dropping out at q(l + 1, hl+2 , s0 ), with a positive probability of
earning strictly less than he would have earned if he waited until that price level.) Therefore,
pi (l + 1, hl+2 , s0 ) ≥ q(l + 1, hl+2 , s0 ) > q(l + 1, hl+2 , s0 ) ≥ bl+2 . Let us show that it would be
strictly better for player i with type s0 to drop out at q(l + 1, hl+2 , s0 ) instead of waiting until
pi (l + 1, hl+2 , s0 ). Indeed, if nobody else drops out in between, or someone drops out before
q(l + 1, hl+2 , s0 ), these strategies would result in identical payoffs. Otherwise payoffs are different,
and this happens with positive probability. Under the former strategy, player i earns
αl+1 (s0 − bl+2 ).
Under the latter strategy, he earns
αl (s0 − bl+1 ),
where bl+1 is the price at which somebody else dropped out. (The probability of getting a spot
m < l is zero by construction.) With probability 1, bl+1 ≥ q(l + 1, hl+2 , s0 ), and with positive
probability bl+1 > q(l + 1, hl+2 , s0 ), so the expected payoff from waiting until pi (l + 1, hl+2 , s0 ) is
strictly less than the expected payoff from dropping out at q(l + 1, hl+2 , s0 ): E[αl (s0 − bl+1 )] <
αl (s0 − q(l + 1, hl+2 , s0 )) = αl (s0 − (s0 −

αl+1
αl (s0 − bl+2 ))) = αl+1 (s0 − bl+2 ).

Therefore, in the subgame following the drop-out of some other player at pi (k, h, s0 ), player i
of type s0 gets position k − 1 with probability 1, and therefore his payoff is αk−1 (s0 − pi (k, h, s0 )).
Now suppose player i dropped out at a price pi (k, h, s0 ) − [U+000F control character] instead of waiting until pi (k, h, s0 ).
If somebody else drops out before pi (k, h, s0 ) − [U+000F control character] or after pi (k, h, s0 ), or drops out at pi (k, h,
↪  s0 )
but player i is chosen to drop out first, then these two strategies result in identical payoffs. The
probability that somebody drops out in the interval (pi (k, h, s0 ) − [U+000F control character], pi (k, h, s0 )) goes to zero as
↪  [U+000F control character]
goes to zero, and the possible difference in the payoffs is finite, so the difference in payoffs due
to this contingency goes to zero as [U+000F control character] goes to zero. Finally, there is a positive probability that
somebody else drops out at pi (k, h, s0 ) and is chosen to drop out first. If player i drops out before
that, at pi (k, h, s0 ) − [U+000F control character], his payoff is αk (s0 − bk+1 ). If he waits until pi (k, h, s0 ), we know
↪  that in the
subsequent subgame his payoff is αk−1 (s0 − pi (k, h, s0 )) < αk−1 (s0 − q(k, h, s0 )) = αk (s0 − bk+1 ).

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[form-feed in source extraction]
Therefore, for a sufficiently small [U+000F control character], it is strictly better for player i with value s0 to drop out at
pi (k, h, s0 ) − [U+000F control character] instead of waiting until pi (k, h, s0 ), which contradicts the assumption that {pj
↪  (·, ·, ·)}
is an equilibrium.
Step 1(b). Now suppose there is mass zero of types of other players dropping out at pi (k, h, s0 ) =
pmax (k, h, s), but there is a positive mass dropping out before pi (k, h, s0 ). Consider a sequence of
small positive numbers ([U+000F control character]n ) converging to 0 as n → ∞ and sequences (πn1 ), (πn2 ), . . . , (πnk−1 ),
↪  where
πnl is the probability that player i with value s0 will end up in position l if another player drops
out at price (pi (k, h, s0 ) − [U+000F control character]n ). Let B = α1 s0 , i.e., the maximum payoff that a player with value s0
can possibly get in the auction. Now, if (πnk−1 ) converges to 1 and (πnl ) converges to zero for all
l < k − 1, then, by an argument similar to the one at the end of Step 1(a), it is better for player
i of type s0 to drop out at some time pi (k, h, s0 ) − [U+000F control character].24 If (πnk−1 ) does not converge to 1, take the
highest l for which (πnl ) does not converge to zero, and take a subsequence of [U+000F control character]n along which (πnl )
converges to some positive number ρ. Let s1 be the value such that for a random draw of types of
remaining players other than i, conditional on each draw being greater than s0 , the probability that
Q
1−F (s )
at least one type is less than s1 is equal to ρ2 (i.e., j6=i 1−Fjj (s01 ) = 1 − ρ2 ). Clearly, s1 > s0 . Take a
small [U+000F control character]n , and consider a subgame following some continuation of history h, hl+2 , where the (l + 2)nd
player drops out at bl+2 , l + 1 players, including player i, remain, and player i gets position l with
```

probability close to ρ (and any position higher than l with probability close to zero). Consider pi (l + 1, hl+2 , s0). There must exist a player, j, such that pj (l + 1, hl+2 , s1) ≤ pi (l + 1, hl+2 , s0). (Otherwise, the probability of player i surviving until position l is less than or close to ρ2 , and thus cannot be close to ρ.) But then, by an argument similar to the one in Step 1(a), in this subgame it is strictly better for player i to drop out slightly earlier than pi (l + 1, hl+2 , s0): Conditional on somebody else dropping out in between, the benefit is close to zero (the probability of getting a position better than l times the highest possible benefit B), while the cost is close to a positive number (the payoff from being in position l + 1 at price bl+2 vs. the payoff from being in position l at price at least s1 −

αl+1
αl (s1 − bl+2)—contradiction.

Expanded PaperInterface specification

```
∀ {clickThroughRate lastDropout value : ℕ → ℝ} {state : EconCSLib.Auction.PaperTheorem8GeneralizedEnglishAuctionState ℕ}
  {rank : ℕ},
  0 < clickThroughRate rank →
    EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout value rank <
        state.clockPrice →
      clickThroughRate rank * (value (rank + 1) - state.clockPrice) <
        clickThroughRate (rank + 1) * (value (rank + 1) - lastDropout rank)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_q_step1_dropping_after_q_review
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 801-823

Verbatim source input

```
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can “swap” such players’ strategies, and therefore there exists an “observationally equivalent”
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
(s − bk+1 ).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set
```

Expanded PaperInterface specification

```
∀ {clickThroughRate lastDropout value : ℕ → ℝ} {rank : ℕ},
  0 < clickThroughRate rank →
    0 ≤ clickThroughRate (rank + 1) →
      clickThroughRate (rank + 1) ≤ clickThroughRate rank →
        lastDropout rank ≤ value (rank + 1) →
          lastDropout rank ≤
              EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout
                value rank ∧
            EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout value
                rank ≤
              value (rank + 1)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_q_mem_interval_review
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 801-823

Verbatim source input

```
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can “swap” such players’ strategies, and therefore there exists an “observationally equivalent”
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
(s − bk+1 ).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set

[Next verbatim source excerpt]

ad was placed in position i is αi . The value per click to bidder k is sk . Bidders are risk-neutral,
and bidder k’s payoff from being in position i is equal to αi sk minus his payments to the search
engine. Note that these assumptions imply that the number of times a particular position is clicked
does not depend on the ads in this and other positions, and also that an advertiser’s value per click
does not depend on the position in which its ad is displayed. Without loss of generality, positions
are labeled in a descending order: for any j and k such that j < k, we have αj > αk .
```

Expanded PaperInterface specification

```
∀ {clickThroughRate lastDropout value : ℕ → ℝ} {rank : ℕ},
  0 < clickThroughRate rank →
    0 < clickThroughRate (rank + 1) →
      clickThroughRate (rank + 1) < clickThroughRate rank →
        lastDropout rank < value (rank + 1) →
          lastDropout rank <
              EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout
                value rank ∧
            EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout value
                rank <
              value (rank + 1)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_q_strict_mem_interval_review
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 528-546

Verbatim source input

```
K 6= N + 1 require only minor modifications in the proof.) Click-through rates αn are commonly
known, with αN +1 ≡ 0. Bidders' per-click valuations si are drawn from a continuous distribution
F (·) on [0; +∞) with a continuous density function f (·) that is positive everywhere on (0, +∞).
Each advertiser knows his valuation and the distribution of other advertisers' valuations.
The strategy of an advertiser assigns the choice of dropping out or not for any history of the
game, given that the advertiser has not previously dropped out. In other words, the strategy can be
represented as a function pi (k, h, si ), where si is the value per click of bidder i, pi is the price at which
he drops out, k is the number of bidders remaining (including bidder i), and h = (bk+1 , . . . , bK ) is
the history of prices at which bidders K, K − 1, . . . , k + 1 have dropped out. (As a result, the
price that bidder i would have to pay per click if he dropped out next is equal to bk+1 , unless the
history is empty, in which case we say that bk+1 ≡ 0.)
Theorem 8 In the unique perfect Bayesian equilibrium of the generalized English auction with
k
strategies continuous in si , an advertiser with value si drops out at price pi (k, h, si ) = si − ααk−1
(si −

bk+1 ). In this equilibrium, each advertiser's resulting position and payoff are the same as in the
dominant-strategy equilibrium of the game induced by VCG. This equilibrium is ex post: the strategy
of each bidder is a best response to other bidders' strategies regardless of their realized values.

[Next verbatim source excerpt]


[form-feed in source extraction]
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can "swap" such players' strategies, and therefore there exists an "observationally equivalent"
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
(s − bk+1 ).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set
```

Expanded PaperInterface specification

```
∀ (clickThroughRate lastDropout value : ℕ → ℝ) (rank : ℕ),
  Continuous fun bidderValue =>
    EconCSLib.Auction.paper_theorem8_generalized_english_ranked_dropout_price clickThroughRate lastDropout
      (Function.update value (rank + 1) bidderValue) rank
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_q_continuous_value_review
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 539-571

Verbatim source input

```
Theorem 8 In the unique perfect Bayesian equilibrium of the generalized English auction with
k
strategies continuous in si , an advertiser with value si drops out at price pi (k, h, si ) = si − ααk−1
(si −

bk+1 ). In this equilibrium, each advertiser’s resulting position and payoff are the same as in the
dominant-strategy equilibrium of the game induced by VCG. This equilibrium is ex post: the strategy
of each bidder is a best response to other bidders’ strategies regardless of their realized values.
The intuition of the proof is as follows. First, with k players remaining and the next highest
bid equal to bk+1 , it is a dominated strategy for a player with value s to drop out before price p
reaches the level at which he is indifferent between getting position k and paying bk+1 per click and
getting position k − 1 and paying p per click. Next, if for some set of types it is not optimal to drop
out at this “borderline” price level, we can consider the lowest such type, and then once the clock
reaches this price level, a player of this type will know that he has the lowest per-click value of the
18

If several players drop out simultaneously, one of them is chosen randomly. Whenever a player drops out, the
clock is stopped, and other players are also allowed to drop out; again, if several ones want to drop out, one is chosen
randomly. If several players end up dropping at the same price, the first one to drop out is placed in the lowest
position of the still available ones, the next one—to the position right above that, and so on.
19
Without this restriction, multiple equilibria exist, even in the simplest English auction with two bidders and
one object. For example, suppose there is one object for sale and two bidders with independent private values for
this object distributed exponentially on [0, ∞). Consider the following pair of strategies. If a bidder’s value is in
the interval [0, 1] or in the interval [2, ∞), he drops out when the clock reaches his value. If bidder 1’s value is in
the interval (1, 2), he drops out at 1, and if bidder 2’s value is in the interval (1, 2), he drops out at 2. This pair of
strategies, together with appropriate beliefs, forms a perfect Bayesian equilibrium.

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[form-feed in source extraction]
remaining players. But then he will also know that the other remaining players will only drop out
at price levels at which he will find it unprofitable to compete with them for the higher positions.
The result of Theorem 8 resembles the classic result on the equivalence of the English auction

[Next verbatim source excerpt]


[form-feed in source extraction]
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can “swap” such players’ strategies, and therefore there exists an “observationally equivalent”
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
(s − bk+1 ).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set
bk+1 = 0.) We will now show that for any i, k, h, and si , pi (k, h, si ) = q(k, h, si ).
Suppose that is not the case, and take the largest k for which there exist such history h (with
the last player dropping out at bk+1 ), player i, and type si (surviving with positive probability on
the equilibrium path) that pi (k, h, si ) 6= q(k, h, si ). Since by assumption, all strategies up to this
stage were pi (·, ·, ·) = q(·, ·, ·), we know that there exists a value smin ≥ bk+1 , such that all players
with values less than smin have dropped out, and all players with values greater than smin are still
in the auction.
Step 1. Suppose for some type s ≥ smin , pmax (k, h, s) ≡ maxi pi (k, h, s) > q(k, h, s). Let s0
be the smallest type, and let i be the corresponding player, such that pi (k, h, s0 ) = pmax (k, h, s).
Without loss of generality, we can assume that there is a positive mass of types of other players
dropping out at or before pi (k, h, s0 ).23
Step 1(a). Suppose first that there is a positive mass of types of other players dropping out
at pi (k, h, s0 ) = pmax (k, h, s). That implies that with positive probability, player i of type s0 will
remain in the subgame following the drop-out of some other player at pi (k, h, s0 ) (since ties are
broken randomly). Let us show that in this subgame, player i of type s0 will be the first player to
23

Otherwise, we have ∀j 6= i, pj (k, h, s0 ) ≤ pi (k, h, s) ≤ pmax (k, h, s) = pi (k, h, s0 ) ⇒ ∀j 6= i, pj (k, h, s0 ) =
pi (k, h, s0 ) and ∀s0 > s0 , pj (k, h, s0 ) > pi (k, h, s0 ). But we also have pi (k, h, s0 ) = pmax (k, h, s) > q(k, h, s) ≥
q(k, h, s0 ), and so for some s0 > s0 we have pmax (k, h, s0 ) > q(k, h, s0 ) and pmax (k, h, s0 ) > pmax (k, h, s). We
can then consider s00 in place of s0 , where s00 is the smallest type, and i0 is the corresponding player, such that
pi0 (k, h, s00 ) = pmax (k, h, s0 ). There is a positive mass of types of other players dropping out before pi0 (k, h, s00 ).

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```

[form-feed in source extraction]
drop out with probability 1. Suppose that is not the case, and let l < k − 1 be the smallest number
such that he gets position l with positive probability.
Consider any continuation of history h, hl+2 , such that the last player to drop out in that history
gets position l+2 and drops out at price bl+2 , player i of type s0 is one of the remaining l+1 players,
there is a positive probability that player i gets position l in the continuation subgame following
history hl+2 , and there is zero probability that player i gets position m for any m < l. Note
that s0 ≥ bl+2 —otherwise it would have been optimal for player i to drop out earlier. Consider
pi (l + 1, hl+2 , s0). There must be a positive mass of types of other players who drop out no
later than pi (l + 1, hl+2 , s0). Take the highest such type, s0 , and the corresponding player j.
It has to be the case that s0 > s0 ≥ bl+2 . It also has to be the case that q(l + 1, hl+2 , s0) is
less than or equal to pi (l + 1, hl+2 , s0). (Otherwise player j with value s0 would be playing a
strategy weakly dominated by dropping out at q(l + 1, hl+2 , s0), with a positive probability of
earning strictly less than he would have earned if he waited until that price level.) Therefore,
pi (l + 1, hl+2 , s0) ≥ q(l + 1, hl+2 , s0) > q(l + 1, hl+2 , s0) ≥ bl+2 . Let us show that it would be
strictly better for player i with type s0 to drop out at q(l + 1, hl+2 , s0) instead of waiting until
pi (l + 1, hl+2 , s0). Indeed, if nobody else drops out in between, or someone drops out before
q(l + 1, hl+2 , s0), these strategies would result in identical payoffs. Otherwise payoffs are different,
and this happens with positive probability. Under the former strategy, player i earns
αl+1 (s0 − bl+2).
Under the latter strategy, he earns
αl (s0 − bl+1),
where bl+1 is the price at which somebody else dropped out. (The probability of getting a spot
m < l is zero by construction.) With probability 1, bl+1 ≥ q(l + 1, hl+2 , s0), and with positive
probability bl+1 > q(l + 1, hl+2 , s0), so the expected payoff from waiting until pi (l + 1, hl+2 , s0) is
strictly less than the expected payoff from dropping out at q(l + 1, hl+2 , s0): E[αl (s0 − bl+1)] <
αl (s0 − q(l + 1, hl+2 , s0)) = αl (s0 − (s0 −

αl+1
αl (s0 − bl+2))) = αl+1 (s0 − bl+2).

Therefore, in the subgame following the drop-out of some other player at pi (k, h, s0), player i
of type s0 gets position k − 1 with probability 1, and therefore his payoff is αk−1 (s0 − pi (k, h, s0)).
Now suppose player i dropped out at a price pi (k, h, s0) − [U+000F control character] instead of waiting until pi (k, h, s0).
If somebody else drops out before pi (k, h, s0) − [U+000F control character] or after pi (k, h, s0), or drops out at pi (k, h,
↪ s0)
but player i is chosen to drop out first, then these two strategies result in identical payoffs. The
probability that somebody drops out in the interval (pi (k, h, s0) − [U+000F control character], pi (k, h, s0)) goes to zero as
↪ [U+000F control character]
goes to zero, and the possible difference in the payoffs is finite, so the difference in payoffs due
to this contingency goes to zero as [U+000F control character] goes to zero. Finally, there is a positive probability that
somebody else drops out at pi (k, h, s0) and is chosen to drop out first. If player i drops out before
that, at pi (k, h, s0) − [U+000F control character], his payoff is αk (s0 − bk+1). If he waits until pi (k, h, s0), we know
↪ that in the
subsequent subgame his payoff is αk−1 (s0 − pi (k, h, s0)) < αk−1 (s0 − q(k, h, s0)) = αk (s0 − bk+1).

19

[form-feed in source extraction]
Therefore, for a sufficiently small [U+000F control character], it is strictly better for player i with value s0 to drop out at
pi (k, h, s0) − [U+000F control character] instead of waiting until pi (k, h, s0), which contradicts the assumption that {pj
↪ (·, ·, ·)}
is an equilibrium.
Step 1(b). Now suppose there is mass zero of types of other players dropping out at pi (k, h, s0) =
pmax (k, h, s), but there is a positive mass dropping out before pi (k, h, s0). Consider a sequence of
small positive numbers ([U+000F control character]n) converging to 0 as n → ∞ and sequences (πn1), (πn2), . . . , (πnk−1),
↪ where
πnl is the probability that player i with value s0 will end up in position l if another player drops
out at price (pi (k, h, s0) − [U+000F control character]n). Let B = α1 s0 , i.e., the maximum payoff that a player with value s0
can possibly get in the auction. Now, if (πnk−1) converges to 1 and (πnl) converges to zero for all
l < k − 1, then, by an argument similar to the one at the end of Step 1(a), it is better for player
i of type s0 to drop out at some time pi (k, h, s0) − [U+000F control character].24 If (πnk−1) does not converge to 1, take the
highest l for which (πnl) does not converge to zero, and take a subsequence of [U+000F control character]n along which (πnl)
converges to some positive number ρ. Let s1 be the value such that for a random draw of types of
remaining players other than i, conditional on each draw being greater than s0 , the probability that
Q
1−F (s)
at least one type is less than s1 is equal to ρ2 (i.e., j6=i 1−Fjj (s01) = 1 − ρ2). Clearly, s1 > s0 . Take a
small [U+000F control character]n , and consider a subgame following some continuation of history h, hl+2 , where the (l + 2)nd
player drops out at bl+2 , l + 1 players, including player i, remain, and player i gets position l with
probability close to ρ (and any position higher than l with probability close to zero). Consider
pi (l + 1, hl+2 , s0). There must exist a player, j, such that pj (l + 1, hl+2 , s1) ≤ pi (l + 1, hl+2 , s0).
(Otherwise, the probability of player i surviving until position l is less than or close to ρ2 , and thus
cannot be close to ρ.) But then, by an argument similar to the one in Step 1(a), in this subgame
it is strictly better for player i to drop out slightly earlier than pi (l + 1, hl+2 , s0): Conditional on
somebody else dropping out in between, the benefit is close to zero (the probability of getting a
position better than l times the highest possible benefit B), while the cost is close to a positive
number (the payoff from being in position l + 1 at price bl+2 vs. the payoff from being in position
l at price at least s1 −

αl+1
αl (s1 − bl+2)—contradiction.

Step 2. In Step 1, we have shown that pmax (k, h, s) ≡ maxi pi (k, h, s) cannot be greater than
q(k, h, s), and therefore for any player i and type s ≥ smin , pi (k, h, s) ≤ q(k, h, s). Take some value
s > smin and player i. Suppose pi (k, h, s) < q(k, h, s). Take some other player j. From Step 1,
we have pj (k, h, smin) ≤ q(k, h, smin) < q(k, h, s), and therefore if player i waited until q(k, h, s)
instead of dropping out at pi (k, h, s), the probability that someone dropped out in between would
be positive, and hence the payoff would be strictly greater (by the definition of function q(), player
i with value s strictly prefers being in position k − 1 or higher at any price less than q(k, h, s) to
being in position k at price bk+1), which is impossible in equilibrium. Hence, pi (k, h, s) = q(k, h, s)
for all s > smin . By continuity, we also have pi (k, h, smin) = q(k, h, smin).

Expanded PaperInterface specification

```
∀ (clickThroughRate : ℕ → ℝ) (boundary : ℕ → ℝ → ℝ),
  (∀ (rank : ℕ), 0 < clickThroughRate rank) →
    have namedStrategy := EOS07GSP.PaperInterface.theorem8ContinuousSourceStrategy clickThroughRate;
    namedStrategy.ContinuousInValuation ∧
      EOS07GSP.PaperInterface.Theorem8ContinuousSourceOneStepBestResponse namedStrategy clickThroughRate ∧
        ∀ (strategy : EOS07GSP.PaperInterface.Theorem8ContinuousSourceStrategy),
          strategy.ContinuousInValuation →
            EOS07GSP.PaperInterface.Theorem8ContinuousSourceOneStepBestResponse strategy clickThroughRate →
              strategy.SupportEq namedStrategy boundary
```

Lean proof endpoint (built separately)

`EOS07GSP.PaperInterface.theorem8_continuous_source_local_best_response_support_unique_review`

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 528-546

Verbatim source input

```
K 6= N + 1 require only minor modifications in the proof.) Click-through rates αn are commonly
known, with αN +1 ≡ 0. Bidders' per-click valuations si are drawn from a continuous distribution
F (·) on [0; +∞) with a continuous density function f (·) that is positive everywhere on (0, +∞).
Each advertiser knows his valuation and the distribution of other advertisers' valuations.
The strategy of an advertiser assigns the choice of dropping out or not for any history of the
game, given that the advertiser has not previously dropped out. In other words, the strategy can be
represented as a function pi (k, h, si ), where si is the value per click of bidder i, pi is the price at which
he drops out, k is the number of bidders remaining (including bidder i), and h = (bk+1 , . . . , bK ) is
the history of prices at which bidders K, K − 1, . . . , k + 1 have dropped out. (As a result, the
price that bidder i would have to pay per click if he dropped out next is equal to bk+1 , unless the
history is empty, in which case we say that bk+1 ≡ 0.)
Theorem 8 In the unique perfect Bayesian equilibrium of the generalized English auction with
k
strategies continuous in si , an advertiser with value si drops out at price pi (k, h, si ) = si − ααk−1
(si −

bk+1 ). In this equilibrium, each advertiser's resulting position and payoff are the same as in the
dominant-strategy equilibrium of the game induced by VCG. This equilibrium is ex post: the strategy
of each bidder is a best response to other bidders' strategies regardless of their realized values.

[Next verbatim source excerpt]

The equilibrium described in Theorem 8 is an ex post equilibrium. As long as all bidders other
than bidder i follow the equilibrium strategy described in Theorem 8, it is a best response for bidder
i to follow his equilibrium strategy, for any realization of other bidders' values. Thus, the outcome
implemented by this mechanism depends only on the realization of bidders' values and does not
depend on bidders' beliefs about each others' types.
Obviously, any dominant strategy solvable game has an ex post equilibrium. However, the
```

Expanded PaperInterface specification

```
∀ (model : EOS07GSP.PaperInterface.theorem8StrictOrderedValueCertificate),
  have localModel :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_exact_schedule_model
      (EOS07GSP.PaperInterface.theorem8StrictOrderedLocalOptimalityCertificateOfStrictValues model);
  EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_local_deviation_sequential_rationality_statement
    localModel.clickThroughRate localModel.value localModel.remaining
    (EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strategy localModel.value localModel.clickThroughRate
      localModel.remaining)
```

Lean proof endpoint (built separately)

```
EOS07GSP.PaperInterface.theorem8_strict_values_ex_post_local_deviation
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 24

Source locator: cited publication, lines 528-590

Verbatim source input

```
K 6= N + 1 require only minor modifications in the proof.) Click-through rates αn are commonly
known, with αN +1 ≡ 0. Bidders' per-click valuations si are drawn from a continuous distribution
F (·) on [0; +∞) with a continuous density function f (·) that is positive everywhere on (0, +∞).
Each advertiser knows his valuation and the distribution of other advertisers' valuations.
The strategy of an advertiser assigns the choice of dropping out or not for any history of the
game, given that the advertiser has not previously dropped out. In other words, the strategy can be
represented as a function pi (k, h, si ), where si is the value per click of bidder i, pi is the price at which
he drops out, k is the number of bidders remaining (including bidder i), and h = (bk+1 , . . . , bK ) is
the history of prices at which bidders K, K − 1, . . . , k + 1 have dropped out. (As a result, the
price that bidder i would have to pay per click if he dropped out next is equal to bk+1 , unless the
history is empty, in which case we say that bk+1 ≡ 0.)
Theorem 8 In the unique perfect Bayesian equilibrium of the generalized English auction with
k
strategies continuous in si , an advertiser with value si drops out at price pi (k, h, si ) = si − ααk−1
(si −

bk+1 ). In this equilibrium, each advertiser's resulting position and payoff are the same as in the
dominant-strategy equilibrium of the game induced by VCG. This equilibrium is ex post: the strategy
of each bidder is a best response to other bidders' strategies regardless of their realized values.
The intuition of the proof is as follows. First, with k players remaining and the next highest
bid equal to bk+1 , it is a dominated strategy for a player with value s to drop out before price p
reaches the level at which he is indifferent between getting position k and paying bk+1 per click and
getting position k − 1 and paying p per click. Next, if for some set of types it is not optimal to drop
out at this "borderline" price level, we can consider the lowest such type, and then once the clock
reaches this price level, a player of this type will know that he has the lowest per-click value of the
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If several players drop out simultaneously, one of them is chosen randomly. Whenever a player drops out, the
clock is stopped, and other players are also allowed to drop out; again, if several ones want to drop out, one is chosen
randomly. If several players end up dropping at the same price, the first one to drop out is placed in the lowest
position of the still available ones, the next one—to the position right above that, and so on.
19
Without this restriction, multiple equilibria exist, even in the simplest English auction with two bidders and
one object. For example, suppose there is one object for sale and two bidders with independent private values for
this object distributed exponentially on [0, ∞). Consider the following pair of strategies. If a bidder's value is in
the interval [0, 1] or in the interval [2, ∞), he drops out when the clock reaches his value. If bidder 1's value is in
the interval (1, 2), he drops out at 1, and if bidder 2's value is in the interval (1, 2), he drops out at 2. This pair of
strategies, together with appropriate beliefs, forms a perfect Bayesian equilibrium.

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[form-feed in source extraction]
remaining players. But then he will also know that the other remaining players will only drop out
at price levels at which he will find it unprofitable to compete with them for the higher positions.
The result of Theorem 8 resembles the classic result on the equivalence of the English auction
and the second price sealed-bid auction under private values (Vickrey, 1961). Note, however, that
the intuition is very different: Vickrey's result follows simply from the existence of equilibria in
dominant strategies, whereas in our case such strategies do not exist, and bids do depend on other
player's bids. Also, our result is very different from the revenue equivalence theorem: payoffs in the
generalized English auction coincide with VCG payments for all realizations of values, not only in
expectation; bidders can be asymmetric; distributions of valuations need not be commonly known.
The equilibrium described in Theorem 8 is an ex post equilibrium. As long as all bidders other
than bidder i follow the equilibrium strategy described in Theorem 8, it is a best response for bidder
i to follow his equilibrium strategy, for any realization of other bidders' values. Thus, the outcome
implemented by this mechanism depends only on the realization of bidders' values and does not
depend on bidders' beliefs about each others' types.
Obviously, any dominant strategy solvable game has an ex post equilibrium. However, the
generalized English auction is not dominant strategy solvable. This combination of properties is
quite striking: the equilibrium is unique and efficient, the strategy of each bidder does not depend
on the distribution of other bidders' values, yet bidders do not have dominant strategies.20,21 The
generalized English auction is a particularly interesting example, because it can be viewed as a
model of a mechanism that has "emerged in the wild."

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[Next verbatim source excerpt]



[form-feed in source extraction]
Proof of Theorem 8
First, note that in equilibrium, for any player i, any history h, and any number of remaining players
k, the drop-out price pi (k, h, si ) tends to infinity as si tends to infinity. (Otherwise there would
exist a player for whom it was optimal to deviate from his strategy and stay longer, for a sufficiently
high value s.) Next, take any equilibrium of the generalized English auction. Note that if in this
equilibrium pi (k, h, si ) > pi (k, h, s0i ) for some k, h, i, and types si < s0i , then it has to be the
case that both types si and s0i are indifferent between dropping out at pi (k, h, si ) and pi (k, h, s0i ).
(Otherwise one of them would be able to increase his payoff by mimicking the other.) Consequently,
we can "swap" such players' strategies, and therefore there exists an "observationally equivalent"
equilibrium in which strategies are nondecreasing in types; also, they are still continuous in own
values. Consider this equilibrium profile of strategies pi (k, h, si ).
Let q(k, bk+1 , s) be such a price that a player with value s is indifferent between getting position
k at price bk+1 and position k − 1 at price q(k, bk+1 , s). That is,
αk−1 (s − q(k, bk+1 , s)) = αk (s − bk+1 )
m
q(k, bk+1 , s) = s −

αk
```

(s − bk+1).
αk−1

Slightly abusing notation, let q(k, h, s) = q(k, bk+1 , s), where bk+1 is the last bid at which a player
dropped out in history h. (This player received position k + 1. If history h is empty, we set
bk+1 = 0.) We will now show that for any i, k, h, and si , pi (k, h, si) = q(k, h, si).
Suppose that is not the case, and take the largest k for which there exist such history h (with
the last player dropping out at bk+1), player i, and type si (surviving with positive probability on
the equilibrium path) that pi (k, h, si) 6= q(k, h, si). Since by assumption, all strategies up to this
stage were pi (·, ·, ·) = q(·, ·, ·), we know that there exists a value smin ≥ bk+1 , such that all players
with values less than smin have dropped out, and all players with values greater than smin are still
in the auction.
Step 1. Suppose for some type s ≥ smin , pmax (k, h, s) ≡ maxi pi (k, h, s) > q(k, h, s). Let s0
be the smallest type, and let i be the corresponding player, such that pi (k, h, s0) = pmax (k, h, s).
Without loss of generality, we can assume that there is a positive mass of types of other players
dropping out at or before pi (k, h, s0).23
Step 1(a). Suppose first that there is a positive mass of types of other players dropping out
at pi (k, h, s0) = pmax (k, h, s). That implies that with positive probability, player i of type s0 will
remain in the subgame following the drop-out of some other player at pi (k, h, s0) (since ties are
broken randomly). Let us show that in this subgame, player i of type s0 will be the first player to
23

Otherwise, we have ∀j 6= i, pj (k, h, s0) ≤ pi (k, h, s) ≤ pmax (k, h, s) = pi (k, h, s0) ⇒ ∀j 6= i, pj (k, h, s0) =
pi (k, h, s0) and ∀s0 > s0 , pj (k, h, s0) > pi (k, h, s0). But we also have pi (k, h, s0) = pmax (k, h, s) > q(k, h, s) ≥
q(k, h, s0), and so for some s0 > s0 we have pmax (k, h, s0) > q(k, h, s0) and pmax (k, h, s0) > pmax (k, h, s). We
can then consider s00 in place of s0 , where s00 is the smallest type, and i0 is the corresponding player, such that
pi0 (k, h, s00) = pmax (k, h, s0). There is a positive mass of types of other players dropping out before pi0 (k, h, s00).

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[form-feed in source extraction]
drop out with probability 1. Suppose that is not the case, and let l < k − 1 be the smallest number
such that he gets position l with positive probability.
Consider any continuation of history h, hl+2 , such that the last player to drop out in that history
gets position l+2 and drops out at price bl+2 , player i of type s0 is one of the remaining l+1 players,
there is a positive probability that player i gets position l in the continuation subgame following
history hl+2 , and there is zero probability that player i gets position m for any m < l. Note
that s0 ≥ bl+2 —otherwise it would have been optimal for player i to drop out earlier. Consider
pi (l + 1, hl+2 , s0). There must be a positive mass of types of other players who drop out no
later than pi (l + 1, hl+2 , s0). Take the highest such type, s0 , and the corresponding player j.
It has to be the case that s0 > s0 ≥ bl+2 . It also has to be the case that q(l + 1, hl+2 , s0) is
less than or equal to pi (l + 1, hl+2 , s0). (Otherwise player j with value s0 would be playing a
strategy weakly dominated by dropping out at q(l + 1, hl+2 , s0), with a positive probability of
earning strictly less than he would have earned if he waited until that price level.) Therefore,
pi (l + 1, hl+2 , s0) ≥ q(l + 1, hl+2 , s0) > q(l + 1, hl+2 , s0) ≥ bl+2 . Let us show that it would be
strictly better for player i with type s0 to drop out at q(l + 1, hl+2 , s0) instead of waiting until
pi (l + 1, hl+2 , s0). Indeed, if nobody else drops out in between, or someone drops out before
q(l + 1, hl+2 , s0), these strategies would result in identical payoffs. Otherwise payoffs are different,
and this happens with positive probability. Under the former strategy, player i earns
αl+1 (s0 − bl+2).
Under the latter strategy, he earns
αl (s0 − bl+1),
where bl+1 is the price at which somebody else dropped out. (The probability of getting a spot
m < l is zero by construction.) With probability 1, bl+1 ≥ q(l + 1, hl+2 , s0), and with positive
probability bl+1 > q(l + 1, hl+2 , s0), so the expected payoff from waiting until pi (l + 1, hl+2 , s0) is
strictly less than the expected payoff from dropping out at q(l + 1, hl+2 , s0): E[αl (s0 − bl+1)] <
αl (s0 − q(l + 1, hl+2 , s0)) = αl (s0 − (s0 −

αl+1
αl (s0 − bl+2))) = αl+1 (s0 − bl+2).

Therefore, in the subgame following the drop-out of some other player at pi (k, h, s0), player i
of type s0 gets position k − 1 with probability 1, and therefore his payoff is αk−1 (s0 − pi (k, h, s0)).
Now suppose player i dropped out at a price pi (k, h, s0) − [U+000F control character] instead of waiting until pi (k, h, s0).
If somebody else drops out before pi (k, h, s0) − [U+000F control character] or after pi (k, h, s0), or drops out at pi (k, h,
↪ s0)
but player i is chosen to drop out first, then these two strategies result in identical payoffs. The
probability that somebody drops out in the interval (pi (k, h, s0) − [U+000F control character], pi (k, h, s0)) goes to zero as
↪ [U+000F control character]
goes to zero, and the possible difference in the payoffs is finite, so the difference in payoffs due
to this contingency goes to zero as [U+000F control character] goes to zero. Finally, there is a positive probability that
somebody else drops out at pi (k, h, s0) and is chosen to drop out first. If player i drops out before
that, at pi (k, h, s0) − [U+000F control character], his payoff is αk (s0 − bk+1). If he waits until pi (k, h, s0), we know
↪ that in the
subsequent subgame his payoff is αk−1 (s0 − pi (k, h, s0)) < αk−1 (s0 − q(k, h, s0)) = αk (s0 − bk+1).

19

[form-feed in source extraction]
Therefore, for a sufficiently small [U+000F control character], it is strictly better for player i with value s0 to drop out at
pi (k, h, s0) − [U+000F control character] instead of waiting until pi (k, h, s0), which contradicts the assumption that {pj
↪ (·, ·, ·)}
is an equilibrium.
Step 1(b). Now suppose there is mass zero of types of other players dropping out at pi (k, h, s0) =
pmax (k, h, s), but there is a positive mass dropping out before pi (k, h, s0). Consider a sequence of
small positive numbers ([U+000F control character]n) converging to 0 as n → ∞ and sequences (πn1), (πn2), . . . , (πnk−1),
↪ where
πnl is the probability that player i with value s0 will end up in position l if another player drops
out at price (pi (k, h, s0) − [U+000F control character]n). Let B = α1 s0 , i.e., the maximum payoff that a player with value s0
can possibly get in the auction. Now, if (πnk−1) converges to 1 and (πnl) converges to zero for all
l < k − 1, then, by an argument similar to the one at the end of Step 1(a), it is better for player
i of type s0 to drop out at some time pi (k, h, s0) − [U+000F control character].24 If (πnk−1) does not converge to 1, take the
highest l for which (πnl) does not converge to zero, and take a subsequence of [U+000F control character]n along which (πnl)
converges to some positive number ρ. Let s1 be the value such that for a random draw of types of

```
remaining players other than i, conditional on each draw being greater than s0 , the probability that
Q
1−F (s )
at least one type is less than s1 is equal to ρ2 (i.e., j6=i 1−Fjj (s01 ) = 1 − ρ2 ). Clearly, s1 > s0 . Take a
small [U+000F control character]n , and consider a subgame following some continuation of history h, hl+2 , where the (l + 2)nd
player drops out at bl+2 , l + 1 players, including player i, remain, and player i gets position l with
probability close to ρ (and any position higher than l with probability close to zero). Consider
pi (l + 1, hl+2 , s0 ). There must exist a player, j, such that pj (l + 1, hl+2 , s1 ) ≤ pi (l + 1, hl+2 , s0 ).
(Otherwise, the probability of player i surviving until position l is less than or close to ρ2 , and thus
cannot be close to ρ.) But then, by an argument similar to the one in Step 1(a), in this subgame
it is strictly better for player i to drop out slightly earlier than pi (l + 1, hl+2 , s0 ): Conditional on
somebody else dropping out in between, the benefit is close to zero (the probability of getting a
position better than l times the highest possible benefit B), while the cost is close to a positive
number (the payoff from being in position l + 1 at price bl+2 vs. the payoff from being in position
l at price at least s1 −

αl+1
αl (s1 − bl+2 )—contradiction.

Step 2. In Step 1, we have shown that pmax (k, h, s) ≡ maxi pi (k, h, s) cannot be greater than
q(k, h, s), and therefore for any player i and type s ≥ smin , pi (k, h, s) ≤ q(k, h, s). Take some value
s > smin and player i. Suppose pi (k, h, s) < q(k, h, s). Take some other player j. From Step 1,
we have pj (k, h, smin ) ≤ q(k, h, smin ) < q(k, h, s), and therefore if player i waited until q(k, h, s)
instead of dropping out at pi (k, h, s), the probability that someone dropped out in between would
be positive, and hence the payoff would be strictly greater (by the definition of function q(), player
i with value s strictly prefers being in position k − 1 or higher at any price less than q(k, h, s) to
being in position k at price bk+1 ), which is impossible in equilibrium. Hence, pi (k, h, s) = q(k, h, s)
for all s > smin . By continuity, we also have pi (k, h, smin ) = q(k, h, smin ).

24

If some other player drops out between pi (k, h, s0 ) − [U+000F control character] and pi (k, h, s0 ), the benefit of staying
↪  longer tends to zero
(it is at most B(1 − π k−1 )), while the cost converges to a positive number (the difference between getting position k
at price bk+1 and position k − 1 at price pi (k, h, s0 )).
```

Expanded PaperInterface specification

```
∀ (model : EOS07GSP.PaperInterface.theorem8StrictOrderedValueCertificate) (n : ℕ),
  have game := EOS07GSP.PaperInterface.Theorem8ContinuousGeneralizedEnglishPayoffGame.ofStrictValues model;
  have scheduledRanks :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_price_sorted_fin_schedule
      (EOS07GSP.PaperInterface.theorem8StrictOrderedLocalOptimalityCertificateOfStrictValues model) n;
  have rankSchedule := EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_fin_schedule_ranks scheduledRanks;
  have localModel :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_strict_ordered_local_deviation_exact_schedule_model
      (EOS07GSP.PaperInterface.theorem8StrictOrderedLocalOptimalityCertificateOfStrictValues model);
  have activeRanks := rankSchedule.toFinset;
  have initialState :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_finite_active_exact_record_cold_start_state localModel
      activeRanks;
  have finalState :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_exact_drop_schedule_final_state localModel initialState
      rankSchedule;
  have G :=
    EconCSLib.Auction.paper_theorem8_bstar_ranked_threshold_terminal_record_source_extensive_dynamic_game_of_states
      localModel initialState finalState;
  have continuation := fun k =>
    EOS07GSP.PaperInterface.theorem7BStarBid localModel.value
      (fun j =>
        EconCSLib.Auction.paper_theorem7_ranked_vcg_tail_payment localModel.value localModel.clickThroughRate j
          localModel.remaining)
      localModel.clickThroughRate (k + 2);
  have namedContinuousStrategy := EOS07GSP.PaperInterface.theorem8ContinuousSourceStrategy localModel.clickThroughRate;
  game.PerfectBayesianEquilibrium namedContinuousStrategy ∧
    (∀ (strategy : EOS07GSP.PaperInterface.Theorem8ContinuousSourceStrategy),
        game.PerfectBayesianEquilibrium strategy →
          strategy.SupportEq namedContinuousStrategy
            EOS07GSP.PaperInterface.Theorem8ContinuousGeneralizedEnglishPayoffGame.nonnegativeSupport) ∧
      ∀ (strategy : EOS07GSP.PaperInterface.Theorem8ContinuousSourceStrategy),
        game.PerfectBayesianEquilibrium strategy →
          G.PerfectBayesianEquilibrium (strategy.inducedActionStrategy continuation localModel.value) ∧
            G.outcomeOf (strategy.inducedActionStrategy continuation localModel.value) = G.vcgOutcome ∧
              strategy.ProfileEq namedContinuousStrategy continuation localModel.value
```

Lean proof endpoint (built separately)

`EOS07GSP.PaperInterface.theorem8_continuous_generalized_english_payoff_game_strict_values_main_conclusion_review`

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation